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 SURVEY

Emerging Trends in Wildfire Detection Through the Lens of Computer Vision and Wildfire Emission Quantification: A Comprehensive Survey

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ABSTRACT Wildfires have become increasingly extreme and frequent, caused by climate change and human activities, and posing considerable threats to ecosystems, air quality, and public health. This survey provides a comprehensive review of state-of-the-art computer vision techniques for both wildfire detection and emissions quantification, covering 27 detection studies and 16 emissions studies published between 2021 and 2024. We explore methodologies that include convolutional neural networks (CNNs), transfer learning, You Only Look Once (YOLO), and hybrid deep learning models, leveraging diverse datasets such as satellite imagery, UAV feeds, and thermal infrared data. The surveyed detection methods achieved accuracy rates ranging from 79.4% to 99.9%, with some real-time models achieving over 15 FPS on edge devices. Emission quantification approaches, including those using burn-area maps and fire radiative power (FRP), support near real-time environmental monitoring. Importantly, the accuracy of emission estimates is tightly coupled with upstream detection and segmentation performance, with uncertainties propagating across the detection–emission pipeline. By integrating multispectral imaging and real-time data processing, these methods enhance the speed and accuracy of fire detection, which is vital for effective wildfire management. Outputs from computer vision-based wildfire detection, including burned area, fire perimeter, and fire radiative power, are key inputs for emission models. As a result, uncertainties at the detection stage can propagate into emission estimates and influence air-quality and climate assessments. However, challenges such as high computational demands, environmental variability, and data quality issues remain. Beyond technical performance, the reviewed approaches have direct implications for operational wildfire management, air quality assessment, and policy-relevant decision-making under real-world deployment constraints. The study also highlights future research directions, including optimizing models for scalability, improving sensor fusion techniques, and addressing the computational limitations of current approaches.

INDEX TERMS Wildfire detection, wildfire segmentation, wildfire emission, deep learning for wildfires.

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I. INTRODUCTION

Wildfires are an increasing environmental hazard which affects ecosystems, air quality, and human health. Wildfires affecting the frequency and intensity are directly linked to

climate change, and global warming and varying precipitation patterns raise the risk of fire occurrences. Wildfires emit large quantities of greenhouse gases, methane (CH_4), and particulate matter (PM), aggravating global climate change and worsening air quality, and increasing the levels of carbon dioxide pollution (CO_2) [1]. The inclusion of satellite remote sensing has enhanced the monitoring of wildfire dynamics at a relatively large scale, which enables a better estimation of emissions and an assessment of the risks involved in this field of activity [2]. The recent study carried out by Matthew et al. [3] states that global wildfire emission monitoring highlights the need for high-resolution satellite data to estimate the effects on the atmosphere and accurately forecast the pattern of fire behaviour. Unmanned Aerial Vehicles (UAVs) are low-altitude aircraft fitted with visual sensors that are effective at delivering high-resolution images required in real-time to detect wildfires early enough to respond to them effectively and efficiently [4]. The early detection of wildfires and estimating their emissions will be useful in reducing their effects, making fire management choices and protecting the health of people in the population at risk [5].

Conventional ways of tracking wildfire usually involve on-site information gathering, manual reports on fires and meteorological information. Such practices are constrained by the scope and the timeliness of the data, and can often lead to a response and detection lag due to them being delayed in processing the issues at hand [6]. The scope and the timeliness of such data commonly cause a delay in responses and detection of the issues, as they are slow to process the issues at hand [7]. Such practices are constrained by, but not always, the scope and the timeliness of the data, frequently leading to a delay in responding and detecting the issues in question. The recent developments in computational intelligence and especially on smoke and fire detection have been found to have potential in monitoring processes to be automated thereby improving the speed and accuracy of detection processes in fire and other wildfire scenarios [7]. To mostly address these problems, such technologies as remote sensing such as satellite imagery, drones, and thermal cameras, are introduced, leading to the formation of a large amount of visual data, which can be analysed to identify them beforehand and measure emissions in totality full [8]. In this context, computer vision, a subfield of artificial intelligence (AI), has become a key tool for automating visual data processing, which can be applied to detect fire events without errors and in a minimal amount of time [5].

The latest developments in the field of computer vision which was largely fueled by the application of deep learning (convolutional neural networks (CNN)) have enhanced the detection of wildfires. These models can detect fire hotspots, smoke, and burn areas in satellite and aerial images more quickly and accurately than conventional techniques do and offer a more dependable and efficient method of detection in comparison to traditional techniques used to detect

fire-related data on a satellite or aerial image. Furthermore, hybrid CNN-Transformer networks have been proved to be useful when trying to identify fires in complicated conditions, such as smoked smoke and minor fire sources [9]. Similarly, deep learning architectures, including Long Short-Term Memory (LSTMs) and Bi-directional Long Short-Term Memory (Bi-LSTMs), have enhanced the detection and control of wildfire in areas that have varying environmental factors [10], [11]. Multispectral imaging combined with hyperspectral imaging has enabled computer vision systems to more effectively distinguish smoke than other atmospheric processes, which increases the precision of emissions estimates made by them [12]. Beyond detection accuracy, the computational cost, energy efficiency, and carbon footprint of deep learning models have become increasingly important considerations in wildfire monitoring, particularly in the context of sustainability-focused research. High-capacity deep learning architectures often require energy-intensive training and inference processes, which may limit their practicality for continuous monitoring and large-scale deployment. These concerns are especially relevant for wildfire-prone regions where monitoring systems must operate persistently, often under constrained energy and connectivity conditions. Consequently, recent research has begun to emphasise lightweight architectures, edge-based processing, and energy-aware deployment strategies as essential complements to algorithmic performance.

Computer vision can also be applied in estimating the wildfire emissions in order to have proper management of wildfires. In addition, it is interesting to be aware of the amount and type of emissions released to establish the effect on the environment. Many studies have used multispectral and thermal satellite measurements which are calculated with machine learning and evaluated to identify the concentration of atmospheric pollutants [13]. These procedures offer a comprehensive understanding of the chemical structure and concentrations of pollutants, which is an important clue on how wildfires have a role to play in the air quality of the region and the world at large [14]. Furthermore, the detection outputs derived from deep learning and edge-based systems are increasingly used not only for fire localization but also for estimating emission impacts. By combining computer vision outputs such as burned area and FRP with observational and numerical data, emission models can quantify greenhouse gas and particulate releases. Figure 1 presents this integrated detection–emission framework. In practical deployments, this coupling also implies that uncertainties and biases introduced during the detection stage, including regional data bias and limited transferability across vegetation types or sensing platforms, propagate directly into emission estimates derived from burned area and FRP. Explicitly accounting for this uncertainty propagation is therefore critical for improving the robustness and regional applicability of emission inventories.

Regardless of the advances, there are a few challenges in the application of computer vision methods to monitor

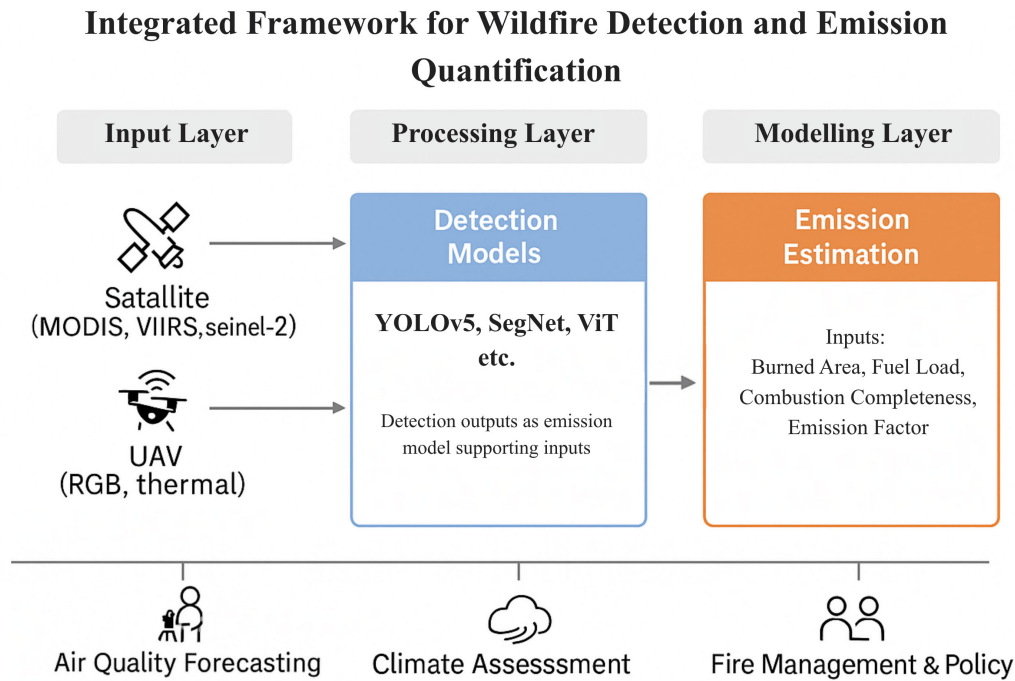


FIGURE 1. The framework illustrates how multi-sensor data are processed using deep learning-based detection models to identify active fires and burned areas. The outputs (burned area, FRP, fire intensity) feed into emission estimation models incorporating fuel load, combustion completeness, and emission factors to derive CO_2 , CH_4 , and $\text{PM}_{2.5}$ emissions for environmental applications.

wildfires. The detection models can be less accurate due to the environmental factors, including thick smoke, cloud cover, and different terrain conditions, in turn, leading to their reduced accuracy [15]. Also, it is computationally difficult to handle large volumes of data in real time, particularly with high-resolution satellite images or UAV footage [16]. Therefore, deep learning models must continue to evolve and become more robust to diverse environmental conditions in order to improve wildfire response strategies [2]. To solve these problems, it is necessary to create more resistant and adaptable models capable of operating under various environmental conditions and effectively analysing large volumes of datasets [1]. Although the wildfire detection and emission estimation are frequently discussed as independent fields of research, they constitute an analytical pipeline in practice. The foundations of emission quantification systems are detection outputs based on computer vision models, namely, fire location, area burned, and intensity. In turn, the reliability of estimated greenhouse gas and particulate emissions is also the direct result of detection accuracy, spatial resolution, and uncertainty. The current review thus takes a more holistic approach and looks at computer vision-based wildfire detection and the implications that the methodology has in terms of estimating emissions through a single framework.

A. SCOPE OF THE REVIEW

Recent reviews have discussed the use of computer vision and machine learning on the problem of detecting wildfires

in broad terms. However, they usually pay attention to particulars, and they are without a holistic approach to the issue of wildfire management. As an example, [17] considered the problem of detecting a wildfire by airborne optical and thermal remote sensing technologies and focused on the hardware of the sensors, which can be hyperspectral or infrared systems on manned and unmanned platforms. Their work, however, does not go as far as to explore the sensor technology, it does not focus on the integrated machine-learning that may be utilised to improve the automated detection services. Similarly, a survey conducted by Mohapatra & Trinh [15] presented a review of early detection technologies of wildfires based on sensor networks, UAVs, and satellite surveillance, but failed to discuss how deep learning models could be used to enhance the accuracy of such technologies. In machine learning, a review of the application of machine learning in wildfire science was explicit and detailed, with focus on fire detection, behaviour prediction, and risk assessment as per the study cited by [13]. Nevertheless, they mainly reviewed literature published earlier than 2020 years and did not pay much attention to the developments in the field of deep learning and real-time data fusion that are required to implement dynamic fire detection. Urbanski et al. [18] on the other hand, were concerned with the emissions made by wildfires and addressed their role in greenhouse gases within the atmosphere. Although informative, this review failed to dwell on how these emissions can be monitored using computer vision or machine learning to reduce them. Recent reviews have also considered deep

learning-based approaches. Bouguettaya et al. [4] mentioned early wildfire detection on UAVs coupled with deep learning algorithms, and the authors indicated the improvements in detecting wildfires in the initial stages. Nonetheless, they focused primarily on UAVs and paid little attention to the inclusion of other sources of data, including ground sensors or satellite images, to create a more robust system. Equally, Ghali et al. [2] reviewed the latest progress in deep learning to classify, detect, and segment wildfires, which classifiers included CNNs and vision transformers. Their research, although it is an improvement, lacked the dimension of the issues of real-time data integration or implementation in dynamic environments. Boroujeni et al. [19] went on to discuss ways AI-enabled UAV systems are used at different phases of wildfire management, such as pre-fire planning and post-fire assessment. Nonetheless, the review mainly discussed UAV capabilities and did not discuss the way other sensor networks can be woven in or find solutions to the issues of deployment in real-life conditions. The intersection of climate change and wildfire events with reviews, including those by Jones et al. [3], and others, have primarily focused on events related to emissions and fire trends caused by climatic changes, including the 2023 wildfires in Canada and Greece. Although these works present viable information on the effects of climatic changes on the frequency of wildfires, they do not address the importance of the use of advanced computer vision methods in the early detection and control of wildfires. The article by Ramos et al. [6] considered the computer vision techniques, especially the deep learning models, such as You Only Look Once v5 (YOLOv5) and Faster R-CNN, to detect wildfires with UAVs and satellite videos. This review, however, was very localised and concentrated only on the detection methods without the implications of wildfires in general, including the emissions and climate change. Finally, the article by Kallimani et al. [20] also discussed the application of Geographical Information Systems (GIS) in real-time forest monitoring and wildfire detection through pattern recognition and image processing to detect wildfires. Nevertheless, the review is still centred on GIS and remote sensing without covering the recent developments in the field of deep learning or computer vision that can potentially increase the accuracy of the wildfire detection systems.

Emissions monitoring and wildfire detection are two aspects of a single monitoring pipeline: real-time detection of active fires can be used to obtain the spatial and temporal mask required to determine the quantity of downstream greenhouse and particulate emissions. In this survey, Section VII shows results of the state of the art in computer vision methods for delineating fire perimeters and hotspots on various sensor platforms. In contrast, Section VIII shows how detection results are converted into burn-area maps and then used with satellite fire radiative power data to estimate emissions in models such as the Global Fire Assimilation System (GFAS). By explicitly treating emission estimation as the subsequent stage of a

vision-based detection workflow, we highlight the end-to-end linkage from algorithmic innovation to environmental impact assessment.

B. CONTRIBUTIONS

This review is a combined work on the latest developments in computer vision and deep learning in the context of wildfire detection and a detailed study of wildfire emissions and their impact on the environment. This review presents a holistic perspective on computer vision object detection and 16 publications published within the same period on wildfire emissions by reviewing 27 publications published between 2021 and 2024. Our survey addresses these areas unlike other reviews which have concentrated on limited parts like sensor technologies or emissions individually, and this is to suggest holistic approaches of enhancing wildfire management. The experience of this review may be a primary contributor to enhancing the early detection systems, real-time surveillance, and mitigation measures, thereby contributing to the minimisation of the negative effects of wildfires on ecosystems and human health.

The review provides:

- 1) A critical analysis of the computer vision tasks of wildfire detection, including image classification, object detection and segmentation methods and even comparing the benefits and limitations of the images used (e.g., satellite, UAV, thermal and multi-spectral images) and discussing their advantages and limitations in various wildfire detection contexts.
- 2) A critical review of the current methodologies, their advantages and disadvantages, the new trends such as multi-modal data fusion, and gaps in the current research environment that need to be filled to maximise the potential of wildfire detection would be provided.
- 3) Further investigation of the effects of wildfire emissions on climate change and air quality, the identification of crucial trends in the factors of emission, and the development of a strategy to reduce the harmful effects of wildfires were discussed in the detailed analysis of the global trends in the emissions of wildfires, relying on 16 studies published in 2021-2024.
- 4) Proposals for future research directions, emphasising integrating advanced deep learning models with real-time data fusion techniques to enhance detection accuracy and monitoring capabilities. This includes recommendations for improving real-time wildfire management systems that respond rapidly to changing conditions, ultimately minimising environmental and economic damage.
- 5) A cross-disciplinary perspective linking computer vision-based wildfire detection with emission quantification, uncertainty propagation, sustainability considerations, and policy-relevant decision-making for wildfire management.

II. STUDY SELECTION METHODOLOGY

In order to come up with a thorough and yet narrow survey, we employed a three-stage systematic procedure to select and locate 27 wildfire detection and 16 emissions articles that are discussed in this paper.

A. LITERATURE SEARCH

We searched IEEE Xplore, Scopus, Web of Science and MDPI using the following Boolean expression:

```
((`wildfire` OR `forest fire`) AND
(`computer vision` OR `deep learning`
```

```
OR `transformers` OR `CNN` OR `YOLO`
OR `emissions`)
```

There were about 450 distinct records brought about by this first search.

B. TITLE AND ABSTRACT SCREENING

We then screened the titles and abstracts independently after eliminating duplicates to exclude:

- Non-vision-based wildfire studies (e.g., purely meteorological models or statistical models)
- Emission studies that do not quantify wildfire carbon or particulate outputs
- Review articles, editorials, and non-peer-reviewed reports.

This step narrowed the pool to 132 candidate papers.

C. FULL-TEXT ELIGIBILITY AND INCLUSION

We retrieved full texts for the remaining 132 and applied the following inclusion criteria:

- Detection studies: Must present a computer vision model (e.g. CNN, YOLO, transformer) for wildfire/fire-smoke detection, with quantitative performance metrics.
- Emission studies: Must provide quantitative estimates of wildfire CO₂ or particulate emissions using remote-sensing or observational inputs (e.g., burn-area, FRP, satellite retrievals).

After consensus discussion, 27 detection studies and 16 emission studies met all criteria and were subsequently categorised by pipeline stage. This transparent selection process enhances reproducibility and ensures that our survey reflects the current state of the art in vision-based wildfire monitoring and emission quantification. The complete workflow of identification, screening, eligibility assessment, and final inclusion as shown in Fig. 2.

III. SYSTEMATIC PROCESSING PIPELINE

We impose a coherent structure on the heterogeneous body of work in wildfire detection based on computer vision and emissions monitoring by organising the surveyed methods into a six-stage processing pipeline. This framework does not only help in explaining how raw data is transformed to actionable emission metrics but also helps in showing how individual studies support, extend or oppose every step in the

workflow. On the pipeline of Fig. 4, the detailed description of a high-level workflow regarding computer vision based Wildfire detection can be expressed as follows: Starting with data acquisition (RGB, thermal, satellite, UAV) by multisensors, followed by preprocessing. Features are then extracted and fed to deep learning detectors (CNN/RNN/LSTM) and the resultant fire maps undergo post-processing to produce real-time alerts and aid in downstream analysis of emissions.

A. MULTI-SENSOR ACQUISITION

Wildfire monitoring uses many types of sensors, each with different strengths and weaknesses in terms of area coverage, image detail, and cost. Ground-based cameras (both RGB and thermal) give clear, fast images of local areas, making them useful for early fire warnings. Satellites such as MODIS and VIIRS, on the other hand, cover the whole world but with lower image detail. Unmanned aerial vehicles (UAVs or drones) occupy the space between the two, as they can cover a large area in a short period and produce images of very high detail. They also map forest structure and fuel levels using LiDAR sometimes. The combination of these tools enables the scientists to monitor fires of various magnitudes, such as small local fires or large ones on a regional level. It has been demonstrated in previous research that ground cameras can enhance early-warning systems in combination with satellite data such as VIIRS, as demonstrated in previous studies by Yandouzi et al. [22] and Ghali et al. [23]. The images provided by UAVs and ground cameras are sharp, which assists in creating and testing detection algorithms, and satellites are necessary to track the fire on a large scale. Local camera-based systems also help validate satellite data and make satellite results easier to understand. The latest deep learning approaches have enhanced the way data from various sensors are integrated. When optical, thermal and radar images are combined, researchers are able to come up with clearer and consistent information on fire activity. This multi-sensor fusion is useful in tracking the changes in the environment. Recent review indicates that data obtained through multiple sources can be used to make wildfire monitoring more effective and reliable [24]. The concepts can inform future systems, which are more interlinked to identify and analyse UAV, satellite, and ground-based data to detect fire.

B. PREPROCESSING AND HARMONIZATION

Raw data retrieved by various sensors should be carefully preprocessed, and then only can they be combined. All of these sensors can have their own distortions or errors and these should be rectified initially. Geometric registration, which brings all images into a common geographic reference, is one of the steps. This operation is used to correct the movement or drift of a platform (e.g., a drone or satellite) of the camera, so that the information of various sensors align correctly on the map. Radiometric calibration removes non-linearities of sensor responses and atmospheric

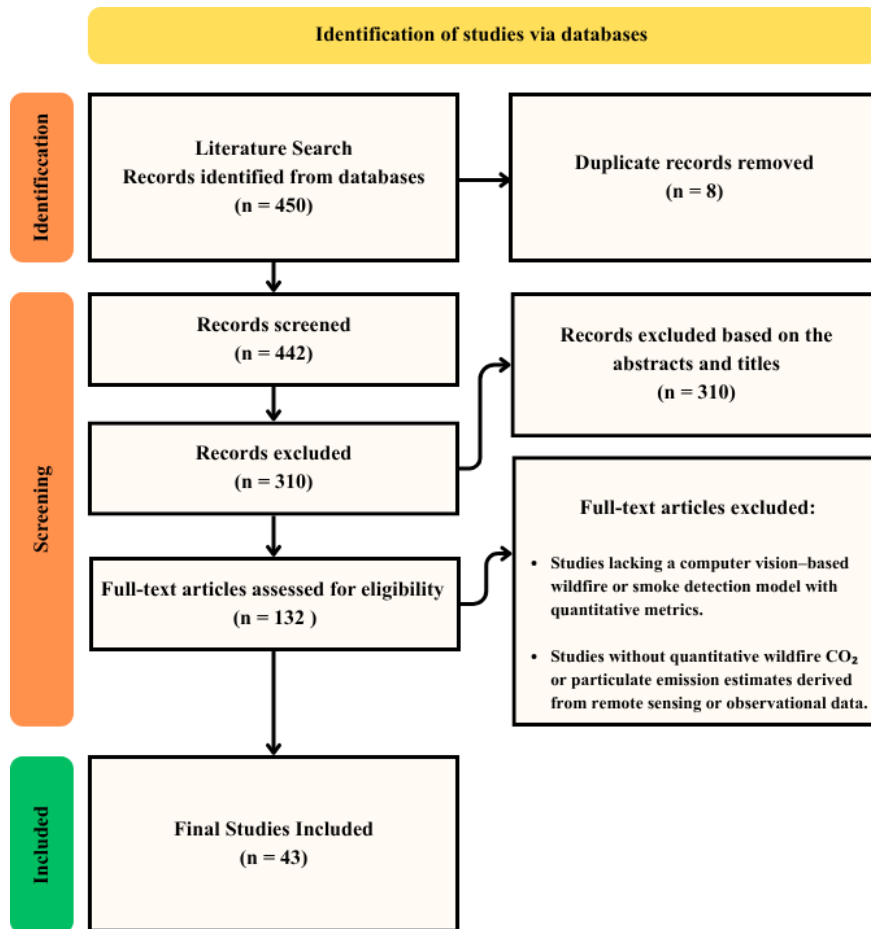


FIGURE 2. PRISMA flow diagram outlining the systematic review process, including identification, screening, eligibility, and final inclusion of studies.

attenuation. The raw data obtained with various sensors should be preprocessed with care before combining them. Both kinds of sensors can possess their distortions or errors and therefore have to be addressed initially. Geometric registration is one of the most important steps to be taken and it aligns all the images to the same geographic reference. This is used to correct the camera motion or drift of the platform (e.g. a drone or satellite) so that the information provided by various sensors aligns properly on the map. Radiometric calibration eliminates sensor response non-linearities and atmospheric attenuation.

Noise reduction via denoising autoencoders or classic filters and data augmentation strategies (e.g., random cropping, spectral jitter) further enhances model robustness to adverse conditions such as haze, smoke occlusion, and variable illumination. Beyond generic denoising, several recent single-image dehazing algorithms offer substantial gains in wildfire contexts by explicitly modelling and removing atmospheric scattering. FrTrGAN [25] introduces a frequency-domain GAN that isolates low-frequency haze components via Fourier transforms before spatial refinement. Advanced filter-based methods leverage Locally Guided

Image Filtering (GGIF) and optimized spatial priors to adaptively smooth fog yet preserve edges. Further, the Haze-CLIP [26] demonstrates that language-guided fine-tuning (via CLIP prompts) can adapt synthetic-trained dehazing nets to real-world conditions with minimal paired data.

C. FEATURE EXTRACTION AND FUSION

At the core of modern detection systems are deep CNNs and YOLO-style one-stage detectors, often augmented with transformer-inspired attention mechanisms to capture long-range contextual cues. Single-modality approaches extract features independently from RGB or thermal streams; however, the recent trend toward cross-modal fusion leverages complementary information, with thermal signatures disambiguating smoke under low visibility and optical textures improving the spatial delineation of burn scars. Fusion techniques surveyed include early concatenation of multi-band inputs, mid-level cross-attention modules that weight feature maps by relevance, and late fusion of independent detection outputs via ensemble learning. Recent high-impact works further advance this fusion paradigm. Long et al. [27] introduce DMPNet, which employs a

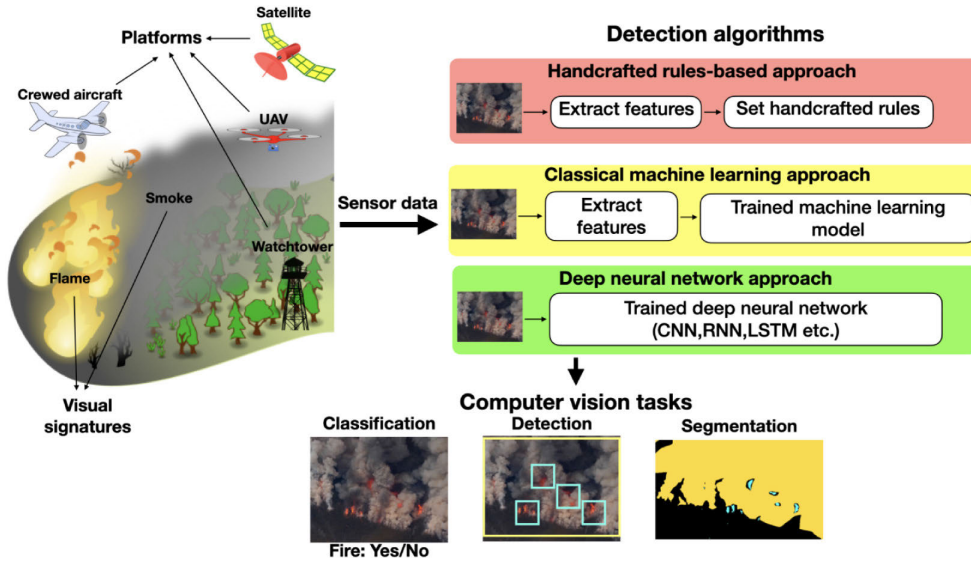


FIGURE 3. Computer vision pipeline for wildfire detection [21].

dynamic scaling fusion (DSF) neck alongside a multi-scale heterogeneous attention (MSHA) module to align adaptively and weight feature maps from UAV-derived RGB and thermal inputs, achieving 86.8% mAP@50 at just 1.2 ms per inference and reducing parameter count by 16.7% relative to YOLOv8n. Elhence et al. [28] propose LCAM-YOLOX, embedding a layer-wise channel attention (LCA) block within the YOLOX backbone and deploying the quantised INT8 model on an FPGA platform; this configuration sustains 195 FPS at 78.1% mAP while consuming only 10.45 W on a Xilinx Kria KV260, demonstrating that attention-driven fusion can be both highly accurate and ultra-efficient at the edge. Together, these advances illustrate the frontier of real-time, multi-sensor fusion for wildfire detection, balancing detection accuracy with the stringent latency and power constraints of deployment platforms.

D. TRAINING, VALIDATION, AND BENCHMARKING

Robust evaluation demands standardised datasets and metrics. Two widely used benchmark corpora, FLAME and BowFire, together with additional custom datasets, provide well-annotated data suitable for both object detection (bounding-box level) and semantic segmentation (pixel-level) tasks. These datasets are of a large variety of environmental environments, including land types and vegetation changes. Their diversity renders them practical in training and testing computer vision to identify wildfires. The vast majority of studies split the data into training, validation, and test sets and provide such common performance measures as precision, recall, F1-score, and Intersection-over-Union (IoU) so that the models could be fairly compared. The performance metric equations are shown in equation 1. Ablation experiments are also carried out by researchers to determine the impact of each component of the model (layers

of attention or fusion blocks) and various preprocessing steps on performance. This assists in making the research more transparent, reproducible and useful in further enhancements.

$$\text{Precision} = \frac{TP}{TP + FP}, \text{Recall} = \frac{TP}{TP + FN},$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1)$$

E. EDGE DEPLOYMENT AND INFERENCE

Moving from lab experiments to real-world wildfire detection systems requires careful tuning of the model's size, speed, and energy use. Smaller and faster versions of models such as YOLOv4/v5 and MobileNet are made with techniques such as pruning, quantisation, and knowledge distillation. These are lightweight models that can operate on small platforms such as NVIDIA Jetson Nano or Xavier or other dedicated hardware to perform fast processing. In reality, they are capable of over 15 frames of drone processing a second (FPS) with a detection latency of less than 100 milliseconds, enabling them to transmit real-time fire alerts. The use of power and heat is also researched by researchers to ensure that these systems can operate continuously in harsh and remote locations.

F. EMISSIONS QUANTIFICATION

In addition to mere fire detection, the interpretation of their environmental effect would be to convert the data on the location and time of fire into the estimated quantity of gases and particles that the fire emits. Systems such as GFAS (Global Fire Assimilation System) use maps of the burnt areas and intensity of the fire. These systems use satellite data on fire brightness (radiative power) and models of the portion of biomass burned to estimate CO₂ and particulate matter emissions. By averaging this data over time, scientists will be

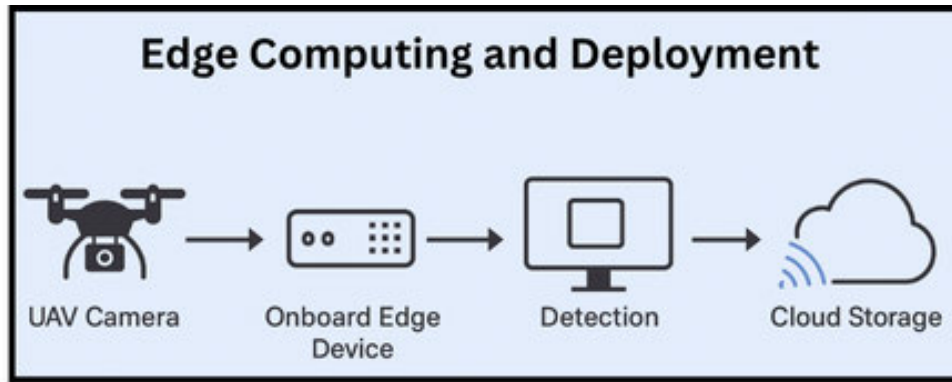


FIGURE 4. Edge computing and deployment flow.

able to estimate the seasonal and annual patterns of emissions, which, in turn, can be compared with the records on the global level, such as approximately 2.4 petagrams of carbon emitted in 2023-2024. It also assists in the research of certain regional trends, including the rise in the number and severity of fires in the Arctic.

Moreover, from the systems perspective, architectural and training choices directly influence the energy consumption and environmental footprint of wildfire detection pipelines. Models with deep backbones, extensive attention mechanisms, or high-resolution inputs may achieve improved accuracy but incur substantially higher computational and energy costs during both training and inference. In contrast, lightweight architectures and edge-oriented designs reduce energy demand by limiting data transmission, shortening inference latency, and avoiding reliance on carbon-intensive cloud infrastructure. These trade-offs highlight the importance of evaluating wildfire detection models not only in terms of accuracy but also with respect to computational efficiency and sustainability, particularly for long-term operational deployment.

IV. COMPUTER VISION TASKS FOR WILDFIRE DETECTION

Recent breakthroughs in computer vision have significantly enhanced the detection of wildfires by enabling the system to automatically detect and analyse fires in real time. Object detection and object segmentation are the two major computer vision tasks that are employed to this end. Object detection assists in identifying the position of a fire in an image or video frame whereas object segmentation identifies the precise shape and size of the fire giving more information on how the fire is spreading. Combined, these methods enable early warning systems to be more precise and dependable in facilitating quicker and more efficient reactions to wildfire incidents.

A. OBJECT DETECTION

Object detection is also one of the areas of computer vision that works on the identification and location of objects in videos or images. It applies deep learning (Convolutional

Neural Networks (CNNs), Region-based CNNs (R-CNNs), and other sophisticated models such as YOLO and Faster R-CNN) to provide real-time detection and localisation services [29], [30]. This has found numerous applications, e.g. autonomous driving, security surveillance and medical diagnostics. In addition, object detection has been the basis of a number of more complex computer vision tasks, such as instance segmentation [31], [32], image captioning [33], and object tracking [34]. Detection of objects may be a single-class or multi-class as shown in Fig. 5. In single-class detection, the system only recognises one type of object (e.g., identifying fire) and puts bounding boxes and confidence scores on it. Multi-class detection detects multiple object types at the same time (e.g., fire and smoke) and provides a more informative and broad overview of the situation to be monitored and analysed.

Object detection in wildfire detection is important to detect fire and smoke in images or live video feeds to instigate earlier and more efficient interventions in a wildfire situation promptly [35]. The contemporary wildfire detector is based on the deep learning models, specifically, on the models of the YOLO (You Only Look Once) type, the main benefit of which is the high detection speed and accuracy. Among them, improved versions of YOLO such as YOLOv8 have demonstrated a high level of performance in the case of fire. As an example, Saydirasulovich et al. [36] constructed a better YOLOv8 model dedicated to UAV imagery by incorporating the BiFormer attention mechanism and the Wise-IoU v3 loss that enhanced localisation accuracy. Their model reached an average precision (AP) of 79.4%, which is also good even in a more complicated forest setup where it is difficult to tell whether it is smoke or a shadow. On the same note, Kim and Muminov [37] refined YOLOv7 and added the Convolutional Block Attention Model (CBAM) to detect tiny smoke areas better. This change boosted the AP by 3.9% to 86.4% and is particularly useful in detecting small-scale smoke early in the development of a wildfire. Another one is Light-YOLOv4 by Wang et al. [38] that is intended to be deployed on an edge. They reached a balance between accuracy and processing speed by integrating MobileNetV3

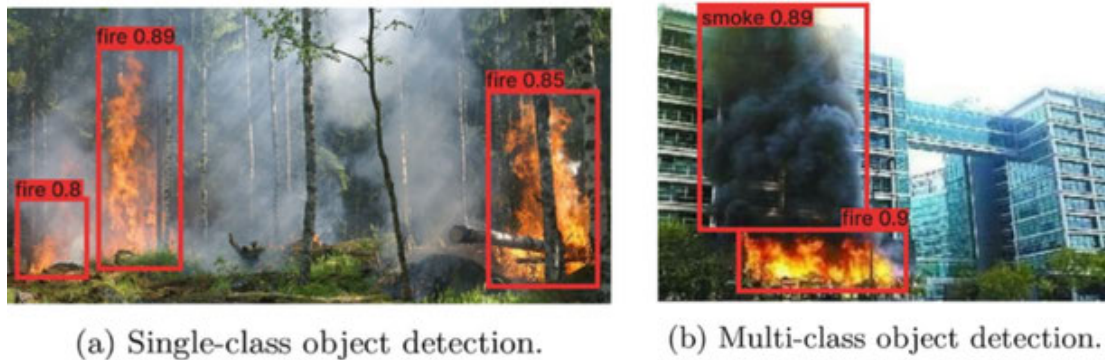


FIGURE 5. Types of object detection. a) Single-class object detection, showing fire as a class. b) Multi-class object detection, showing two classes, fire and smoke [6].

of 85.64% mean Average Precision (mAP) at 71 FPS, which proved to be an optimal balance between accuracy and processing speed in resource-constrained settings. The summary of computer vision-based methods of wildfire detection is demonstrated in Fig. 4.

Further, NVIDIA Jetson Xavier has been heavily utilised as edge devices to enable real-time wildfire detection, which enables processing to be done on-site instead of being shipped to far cloud servers. Indicatively, a smart edge setup was used by Kristiani et al. [39] in detecting flames and smoke by combining YOLO and SSD MobileNetv1 in their study. This allows quick processing the data locally, which will decrease the delays due to cloud transmission and enhance the reliability of the system in remote locations. Equally, Shamtai and Demir [40] presented an embedded system based on the Nvidia Jetson Nano to operate as a UAV-based operations system. Their model had a classification accuracy of 96% and it facilitated continuous aerial surveillance and real-time communication with ground stations, which improved the capability to detect and intervene on the fire at an early stage. In a different experiment, Titu et al. [41] utilised knowledge distillation, where a bigger YOLOv8m model was trained to produce a smaller and more efficient YOLOv8n model that can be utilised in drone platforms. This drone achieved the highest accuracy of 89.23% in field experiments, which indicates high versatility in various terrains and remote locations, which is suitable in the real-world application of wildfire surveillance.

The development of YOLO-style models, attention, and edge computing has transformed the wildfire detection system and made them faster and more accurate regarding detection. Therefore, they can be useful during the real-time management of wildfires. The future studies are expected to streamline these systems to be even more environmentally friendly and resource-wise efficient.

B. IMAGE SEGMENTATION

Image segmentation is one of the fundamental tasks in computer vision, which breaks down an image into separate

parts, each pixel of an image being labelled with a specific label representing that object or area. This may be termed semantic segmentation, wherein every pixel is labelled in an image according to the object type (e.g. tree or fire), or instance segmentation, wherein the individual instances of objects of the identical type are also differentiated. The most popular are Fully Convolutional Networks (FCNs) and U-Net that are extensively applied in medical imaging, autonomous driving, and monitoring the environment [42], [43]. The forest fire image segmentation examples are presented in Fig. 6.

The segmentation of images in the wildfire detection is essential to isolate fire and smoke in different and complicated surroundings to make early detection and tracking. Various deep learning architectures and CNNs to Transformer-based models are used to enhance the accuracy of segmentation, especially in dealing with smoke and flames with irregular shapes and mixed backgrounds. Another of those models is the MsFireD-Net created by Hossain and Zhang [21] which combines a technique called Reversed Depthwise Separable Transposed Convolution (RDSTC). Being much lighter than a UAV, this CNN is optimized to be applied to a UAV, which is better than the accuracy and computational efficiency of the U-Net.

Moreover, the Fire Dynamic Vision (FDV) pipeline by Sages and Quaife, [44], offers a system of multi-scale segmentation and tracking and performs RGB and HSV colour space processing to segment dynamic fire fronts and plumes. The model is very useful in the tracking of fire spread under different space and time variables, which entails its division and statistical processing stages in order to be used at real time. The other significant contribution is by Li et al. [45] with the WSDD-Net, which is a DenseNet based model aimed at minimizing false positives in wildfire smoke segmentation with dilated convolutions and candidate smoke region segmentation system. Balance cross-entropy loss in WSDD-Net also improves the results on complicated wildfire background.

Wang et al. [46] introduced a Swin Transformer model with adaptive multi-scale attention mechanism, that improves the fire edge detection with intersection over union (IoU) of 86.73 percent and precision of 91.23 percent (wang-2024). The model has solved the problem of class imbalance, enhanced the accuracy of segmentation, particularly in determining the irregular edges of flames and smoke. In a very fine aerial image, Perrolas et al. [47] proposed a quad-tree search approach with a CNN to sequentially divide informative regions of the image, hence enhancing computational performance and accuracy of its segmentation with a large aerial image without reducing its segmentation accuracy. Despite their strong ability to capture long-range spatial dependencies, transformer-based and foundation models typically require large volumes of annotated data for effective training, which may not be readily available across diverse wildfire-prone regions. Moreover, the complexity of attention-based mechanisms reduces model explainability, making it difficult to interpret detection outcomes in safety-critical wildfire monitoring scenarios. These models may also exhibit reduced robustness under domain shift, such as changes in vegetation types, atmospheric conditions, or sensing platforms, which can affect generalisation in real-world deployments.

Another method by Xiong and Yan [49] is the support vector machines (SVM) support of smoke detection through super pixel division. Simple Linear Iterative Clustering (SLIC) applied to superpixel segmentation is effective to distinguish between smoke and noise (clouds) and can be therefore used in the early detection of smoke in trees. Together, these models present the innovations in segmentation methods that meet the various problems of wildfire detection and allow precise monitoring of fires in real-time and under different environments.

The progress in computer vision has significantly improved the system of wildfire detection by relying on the methods of object detection and image segmentation to carry out real-time monitoring. Object detection takes place due to deep learning, including YOLO and Faster R-CNN, which is essential to detect fire and smoke on UAV and satellite images quickly, even at limited resources. These models have been optimized to be deployed quickly with recent innovations, including attention mechanisms and edge computing. Image segmentation supplements it with a pixel-level classification, which is required to perform accurate fire lines isolation and to separate smoke and clouds. The U-Net, WSDD-Net, and Swin Transformer techniques improve segmentation accuracy under more complex conditions and can detect irregular fire patterns more effectively. Such developments improve the early warning and proactive management of wildfires, which eventually result in more effective response and mitigation actions.

V. TYPES OF IMAGERY IN WILDFIRE DETECTION

All the forms of imagery are at the centre of the wildfire detection so that the fire events can be properly detected and observed in a timely manner. Each type has its own

benefits and is applicable to certain aspects of detection, so the researchers and responders can choose which ones would be the most effective depending upon the factors, which include area coverage, data cost, data availability, data frequency, data acquisition, detection precision, and environmental conditions. This paragraph summarises the most common forms of imagery used in the detection of wildfires.

A. RGB IMAGES

RGB (Red, Green, Blue) images are the foundation of digital imaging, representing visible light in three colour channels, thus capturing images much like the human eye perceives. In computer vision, RGB images are frequently utilised for tasks involving colour differentiation, object detection, and image segmentation [50]. With the help of deep learning models such the CNN, RGB imagery can be used to extract reliable features based on their colour and spatial characteristics, which is why it is highly useful in applications such as autonomous navigation and facial recognition. The convenience of computation for RGB images in the CNNs is because of its colour and textural content and makes this format a dominant option in many real life applications where clear and immediate visual information is needed [51]. The RGB images depicted in Fig. 7.

RGB images possess wide use in work, which needs high interpretability and real-time analysis. Similarly, recent object detectors and segmenters, such as YOLO and U-Net, are trained to process RGBs, on which there is enough colour contrast and detail to understand the scene. The compatibility of RGB with the representation of natural colours, as well as the fact that it can be easily implemented in diverse applications, including augmented reality and environmental surveillance, is facilitated by the latter feature of the colour model and its popularity among numerous applications and uses [52]. The most crucial in the wildfire monitoring process is RGB images, which can detect the presence of the fire based on the visible indicators of fire (smoke and flames), particularly at the initial stages of the process [53]. The UAVs (Unmanned Aerial Vehicles) with RGB cameras are also especially useful in surveying the smoke and flame indicators at an elevated level and assist in the initial recognition and estimation of fires [54]. The datasets such as the FLAME2 dataset contain RGB imagery taken by drones and this makes the machine learning models more effective in accurately identifying flame zones in various environmental conditions. When object detection algorithms process this RGB data, it enables efficient monitoring and localisation of fires across accessible terrains [53].

RGB imagery is widely used in wildfire detection due to its high spatial resolution and accessibility from standard, low-cost cameras. It effectively supports detailed inspection of fire boundaries, smoke dispersion, and affected landscapes. When combined with thermal imaging, RGB data enhances detection accuracy by providing complementary visual cues

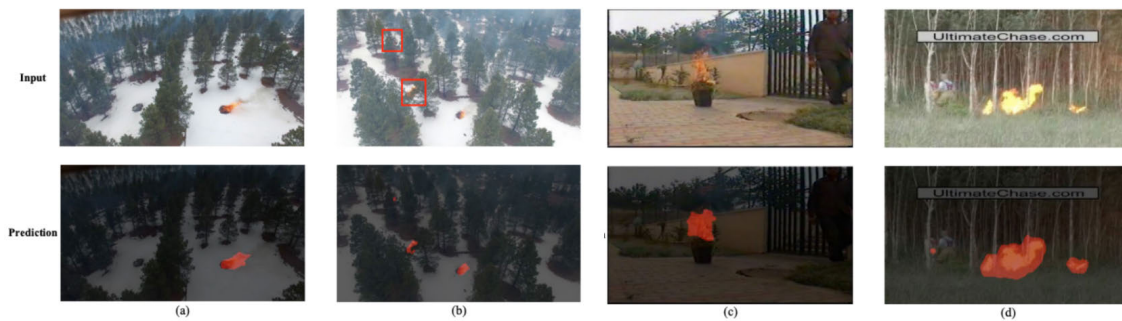


FIGURE 6. Examples of forest fire segmentation [48]. a) The input image and predicted fire-segmented area. b) The red boxes mark some fire areas occluded by trees and predicted fire-segmented areas. c) Input image of a small fire and fire-segmented area. d) Input image of a small forest fire and fire-segmented area.

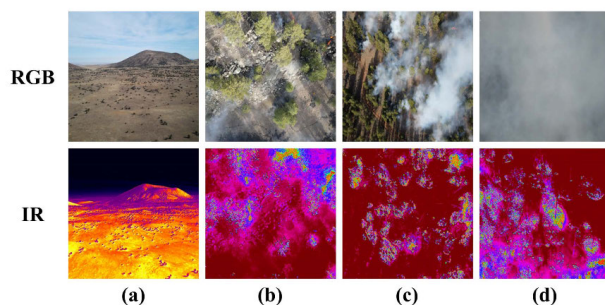


FIGURE 7. RGB (Red, Green, Blue) and Infrared Images examples [45]. a) Smoke-free RGB Image and corresponding Infrared image. b) RGB Image with a slight quantity of smoke and corresponding Infrared image. c) RGB Image with smoke and corresponding Infrared image. d) RGB image full of smoke and corresponding Infrared image.

across the visible and thermal spectra [55]. However, RGB sensors depend on visible light and perform poorly under low visibility, dense smoke, or nighttime conditions, as they cannot capture thermal information to identify hotspots or fire intensity [56].

RGB images can be best utilised in detecting wildfires when visibility is clear and the light is sufficient, and they can best be used in daytime observation of open or semi-forested regions. Use of drones with RGB cameras can provide high-resolution images that can be used to get real-time images of the fire perimeter and monitored smoke. Furthermore, the RGB data enhances infrared and thermal data by providing spatial information, which enhances accuracy of combined wildfire monitoring systems in diverse environmental specifications.

B. THERMAL INFRARED IMAGES

Thermal infrared (IR) imaging captures radiation emitted by objects and enables temperature-based scene interpretation, unlike visible-light imaging, which relies on reflected light. In computer vision, thermal IR sensors are widely used for applications requiring heat signature detection, including wildfire monitoring, particularly under low-visibility or nighttime conditions [56], [57]. Among thermal bands,

mid-wave infrared (MWIR, $3\text{--}5\ \mu\text{m}$) is highly sensitive to the elevated temperatures associated with active flames, while long-wave infrared (LWIR, $8\text{--}14\ \mu\text{m}$) captures lower-temperature gradients useful for contextual and post-fire analysis. These characteristics allow thermal imagery to detect active fires, hotspots, and residual heat that may be invisible to RGB sensors, making it a critical modality for robust wildfire detection and progression monitoring [58]. The Thermal infrared images are shown in Fig. 7.

Thermal IR imaging has now become an important tool for locating and tracking wildfires when visibility is difficult due to smoke, fog, or darkness. Thermal IR sensors are the best in monitoring the spread of wildfires in unfavorable conditions given that the thermal radiation is not affected by the smoke or the intensity of lights. Research has shown that the combination of thermal IR and visible-light sensors allows improving detection accuracy because the thermal IR information visualizes zones of high temperature and hot spots, which is extremely useful in early intervention and continuous monitoring of areas of concern [59]. Thermal IR imaging also enabled more confident segmentation of fire areas and hot spots in dense smoke in the FLAME2 dataset, which also contained RGB and IR data, thus permitting more reliable fire area classification and hot spot detection [53].

Thermal infrared (IR) imagery is highly effective for wildfire monitoring because it operates independently of visible light, enabling fire detection at night or through dense smoke. It can also identify residual heat after flames subside, supporting post-fire assessment and containment analysis [59]. Combining thermal and visible imagery improves situational awareness by integrating spatial and thermal perspectives. However, high-resolution thermal sensors are expensive, may produce false positives from non-fire heat sources, and often lack contextual detail without visible reference imagery [60]. Consequently, thermal IR is typically used alongside RGB data to ensure comprehensive and accurate wildfire monitoring.

Thermal infrared (IR) imagery is considered a very efficient method of detecting wildfires in the situation of low visibility, e.g., during a night shift or smoke-filled

air. Thermal IR sensors installed on satellites and drones are applicable to large-scale observation of temperature, identifying hotspots, and evaluating the level of containment. Thermal IR data is also used to augment multispectral burned-area and vegetation data in post-fire environments to give details of residual heat and surface temperature processes, which can then be used to support early recovery and re-ignition risk analyses.

C. LIDAR IMAGES

Light Detection and Ranging (LiDAR) is a remote sensing method that uses laser light pulses to measure distances between sensor and various surfaces [61]. This technology generates highly accurate 3D point clouds that map terrain features, vegetation structures, and even under-canopy landscapes. In wildfire detection and management, the accurate mapping of LiDAR is invaluable in pre and post-fire surveys because it is applicable in examining the vegetation cover, fuel loads and structural alterations in forest cover. LiDAR facilitates the determination of regions at a high risk of wildfire by creating elevation models and canopy height measurements, which can help in fire prevention efforts in particular situations as well as general fire prevention strategies at large [61]. LiDAR helps determine areas where wildfires are likely to occur by creating height models and canopy height measurements, which in turn can be useful in fire prevention efforts, in both specific and broadly fire prevention efforts in general. The finer details of LiDAR 3D data gives us an in-depth analysis of the topography and vegetation, so that we can understand the fire behaviour in the complex terrain better. LiDAR may be used in the wildfire monitoring process to map the vegetation prior and post fires in order to determine the intensity of burns and recovery of the ecosystems. Combining LiDAR data with multispectral images, the scientists will be able to determine fuel loads and detect structural weaknesses of forests that could affect fire propagation. The fact that LiDAR can penetrate vegetation also allows studying the features of understory, which is essential in terms of building fuel management plans and estimating possible fire lines [62].

LiDAR is an important tool to evaluate the fuel availability, fire behavior modeling, and post-fire analysis because it provides high-resolution, three-dimensional forest structure, terrain, and vegetation density data [63]. Its capability of working under different light conditions enables it to capture the data at night or under heavy cover where the other traditional imaging methods are restricted. The 3D structural data facilitate the fire spread modeling, location of the vulnerable areas and the use of the targeted firefighting tactics. Nevertheless, LiDARs are expensive, data-heavy, and cannot withstand inclementals like rain or fog that disrupt laser interferences [64]. Due to these limitations, LiDAR is frequently used together with multispectral or thermal imaging to create a more detailed wildfire assessment. It is especially applied to produce vegetation and fuel maps in

tricky conditions and post-fire analysis of the severity of burnt areas, biomass loss, and vegetation recovery [61], [65].

LiDAR imaging can aid in the management of wildfires with the provision of high-resolution 3D data on the vegetation structure and terrain, which is useful in preventing wildfires, monitoring them, and recovering them after occurrence. Despite the potential constraints of LiDAR systems including cost and processing capabilities, they supplement other modalities of remote sensing by enhancing knowledge of the fire prone lands and also supplementing effective approaches to managing any wildfire.

D. SATELLITE IMAGES

Satellite imagery has been a vital tool in wildfire detection because it allows monitoring large geographic areas around the clock [66]. High-altitude satellites with multispectral and thermal sensors can monitor the wildfires from the beginning to the end, providing information on the location, intensity, spread, and post-fire impacts. The Geostationary Operational Environmental Satellites (GOES) and the Moderate Resolution Imaging Spectroradiometer (MODIS) that are commonly used for wildfire monitoring allow nearly real-time observations [67]. Also, Sentinel-2 provides high resolution imaging that can be used for detailed analysis of areas affected by fire. Further, thanks to the rapid development of satellite-based sensors, the reliability of fire detection has increased, allowing wildfires to be rapidly and accurately located and studied in remote, inaccessible areas.

Satellite images have a wide range of uses in wildfire monitoring [66]. It is widely used for active fire detection, smoke dispersion monitoring and severity of burnt area assessment. Satellites equipped with multi-spectral and thermal imaging enable the acquisition of information relevant to fires beyond the visible spectrum, especially in shortwave infrared (SWIR) and thermal infrared (TIR) bands, which are very effective for penetration through smoke layers and identification of hotspots [68]. This ability is particularly useful in dense vegetation where visual data alone may not be enough. Various spectral indices such as Normalised Burn Ratio (NBR) are calculated from remote sensing data and these indices are used for determining the burnt area extent and vegetation loss. These indices used before and after fire enable scientists to quantify burn severity and predict recovery potential of ecosystems [69].

Satellite imagery has a large spatial coverage and a high temporal frequency, especially from geostationary satellites, such as GOES, which can have 10-minute updates [67]. The high-altitude perspective enables satellites to see the progress of fires over extensive and remote areas that are not accessible to ground-based instruments for early warning systems and for the sustained monitoring of fires. Further, satellites can compare pre- and post-fire conditions with multi-temporal data, supporting effective recovery planning by evaluating vegetation regrowth and ecosystem impact. Although satellites provide vast coverage,

TABLE 1. Overview of the use cases of different imagery types in wildfire management.

Use Case	RGB	Thermal IR	Satellite	Aerial (UAV)	LiDAR
Early Fire Detection	✓	✓	✓	✓	✓
Nighttime Monitoring	✗	✓	✓	✓	✓
Large-Scale Fire Monitoring	✗	✓	✓	✗	✗
High-Resolution Fire Mapping	✓	✓	✗	✓	✓
Post-Fire Damage Assessment	✓	✓	✓	✓	✓

TABLE 2. Comparison of imagery types in wildfire detection.

Imagery Type	Advantages	Disadvantages	Applications	Example Systems/Datasets
RGB	High availability, cost-effective, high spatial resolution	Limited in low visibility, no thermal data	Daytime fire detection, UAV monitoring	FLAME2, U-Net, YOLO
Thermal IR	Detects heat sources in low visibility, effective at night	Expensive, may produce false positives	Nighttime monitoring, hot spot detection	Himawari-8, FLAME2
Satellite	Wide area coverage, high temporal frequency	Limited by cloud cover, low spatial resolution	Large-scale fire monitoring, post-fire analysis	GOES, MODIS, Sentinel-2
Aerial (UAV)	High spatial resolution, flexible deployment	Limited battery life, weather restrictions	Real-time fire perimeter mapping, small fires detection	DJI drones, FLAME2
LiDAR	Provides 3D terrain data, captures under-canopy details	Expensive, requires heavy processing	Pre-fire fuel load analysis, post-fire burn severity	Airborne LiDAR systems

their effectiveness is sometimes limited by cloud cover, atmospheric disturbances, and relatively low spatial resolution compared to drone or UAV imagery [70]. The fixed revisit intervals of satellites such as MODIS and Sentinel-2, typically 1–2 days, can delay real-time monitoring, especially for rapidly spreading fires [71]. Similarly, while satellite data are effective for broader trends, the lower spatial resolution may miss smaller fires or less intense heat signatures. Satellite data often needs integration with UAV or ground-based sensors to enhance fire monitoring precision, especially for localised fire detection and immediate response needs [72].

The satellite imagery is best applicable to the large-scale wildfires detection and monitoring in the long run, especially in remote or rugged areas. Its constant temporal resolution can be analyzed to determine the patterns of fire expansion, the effects of fires cumulatively and post-fire recovery of the ecosystem in months or years, useful in fire-risk mapping and emergency response planning. Satellite data when used with the high-resolution drone imagery offer a based-in-capacity support to wildfire monitoring needs, both regional and local, and at a multi-scale level.

E. AERIAL IMAGES

The aerial shots are obtained either through aeroplanes, helicopters or unmanned aerial vehicles (UAVs) and offer a very high resolution of images at various flexible altitudes and is therefore able to monitor wildfires in detail and at localised levels. Aerial platforms are more affordable to deploy and provide more spatial resolution and operational capability, enabling it to detect fires early, monitor the fire in operation, and survey the damage after the fire, which is more accurate than satellite imagery in terms of space [70]. These aerial data can consist of RGB, thermal, and multispectral imagery and be utilized in fire detection, mapping the

perimeter, spread analysis, and assessing the burn severity. Specifically, UAVs would be in high-demand in remote or low-visibility zones, where small fires and hotspots can be detected by thermal sensors in close-to-real time. These functions render aerial imaging more useful when it comes to operational management of wildfires as well as post-fire assessments of the natural environment ecology [70].

Small or early fires can be missed by satellites, but aerial imagery can provide high-resolution data to identify them, especially when the UAVs are used [73]. Drones are able to respond quickly, operate safely in fire zones, and deliver real-time and high-precision information to make timely decisions in facing the firefighting process [74]. UAVs with multispectral cameras are able to record information outside of the visible spectrum, which is useful in measuring vegetation health, fuel moisture, and fire-prone regions. Nevertheless, short flight periods, poor weather conditions, and legal limitations limit the use of UAVs [75]. Real time analysis also requires a lot of processing power when handling high-resolution data. The choice of sensing platform introduces important trade-offs between spatial resolution, temporal coverage, and computational cost. Although high-resolution UAV imagery can be useful in localising the location of a fire, the satellite-based data are more comprehensive in terms of space but can create issues of temporal variations, which can influence the detection of a fire and subsequent estimation of emissions. Nevertheless, aerial shots can be of enormous importance in mapping out the boundaries of the fire, tracking its evolution, and performing post-fire analysis in inaccessible locations or those that are remote to visit [76].

The multiplicity of types of imagery improves the accuracy, efficiency, and scalability of the wildfire monitoring systems. The Table 1 highlights the use of RGB, Thermal IR, Satellite, Aerial (UAV), and LiDAR imagery in different

applications. The RGB image is useful in daytime fire detection and mapping a surface in detail whereas Thermal IR is better in low light or smoggy environments where it detects the presence of heat. Satellite imagery offers large scale spatial coverage on large scale surveillance and post-fire assessment and UAVs offer high resolution and flexible imaging on real-time mapping of the fire perimeter. LiDAR supplies additional 3D structural data that can be used in pre-fire fuel analysis and post-fire damage assessment. As Table 2 indicates, technologies have drawbacks, such as RGB does not work well in low visibility, Thermal IR can be expensive, satellites have cloud-cover and resolution problems and UAVs can be limited by flight duration and weather. LiDAR is data-intensive and costly, but delivers unsurpassed structural detail. Combined, these imaging systems complement each other, leading to sound and effective systems of early identification, quick reaction to the disaster, and post-fire recovery. Although wildfire imagery datasets are becoming increasingly available, the issue of data bias is still widespread since a significant part of datasets is geographically localized and serves a particular kind of vegetation, climatic regime, or fire regimes. This is an uneven coverage that may influence model generalisation when extended to areas of different fuel properties or atmospheric conditions. The most effective way to counter these biases is to work with multi-region datasets offering a more diverse set of data that will enhance the transferability of the wildfire detection models.

VI. WIDELY UTILIZED WILDFIRE DATASETS IN RESEARCH

In the fast-moving area of wildfire detection and segmentation, data are simultaneously both the foundation and the challenge for robust computer vision algorithms. High-quality, diverse datasets provide the backbone for creating, testing, and tuning detection algorithms. Literature on wildfire detection showed that, among a great number of datasets that were used for model training and validation, two datasets have been mentioned over and over again because of their strength and usefulness: FLAME, and BowFire. These are some of the general datasets found in many research papers and study articles that we go through, which act as the main database for training, developing, and validating fire and smoke detection algorithms.

A. FLAME

One of the prominent datasets often referenced in wildfire detection research is the FLAME (Fire Luminosity Airborne-based Machine learning Evaluation) dataset. It is developed to address the devastating effects of wildfires on human and animal lives; this dataset is designed to support the prediction of fire behaviour, thereby improving fire management strategies and minimising risks for firefighters. The FLAME dataset was created during a controlled burn in Northern Arizona, USA, using drones (UAVs) as the primary data-collection tools [77].

The main features of the FLAME dataset include Fire Classification and Fire Segmentation. It contains 39,375 labelled frames (either “Fire” or “Non-Fire”) for training and an additional 8,617 for testing fire classification models, with a particular emphasis on NN-based methods. The dataset also contains 2,003 frames with pixel-wise segmentation masks as ground truth. The FLAME dataset is designed with high spatial and temporal detail, combining UAV-based RGB and thermal imagery captured during controlled burns. It includes pixel-wise annotated masks for segmentation and a large number of labelled frames for classification tasks. These design features make it highly suitable for training deep learning models for early fire detection, analysing flame dynamics for behaviour prediction, and assessing risk mitigation strategies under controlled and real-world scenarios. Consequently, this dataset is valuable to researchers and practitioners in wildfire management. Numerous studies, such as [78], [79], and [80], have used this dataset and achieved meaningful results. The Fig 8 shows FLAME dataset fire and no-fire images.



FIGURE 8. Examples from the FLAME dataset. (Top): Fire images; (Bottom): non-fire images [2].

B. BowFIRE

BowFire (Best of Both Worlds Fire Detection) dataset [81] is a publicly accessible dataset that is specifically created with the purpose of supporting fire detection research. It has 226 images, 119 fire images, and 107 non-fire images that have different resolutions. Images of fire cover different emergency events, which include wildfires, burning buildings, car accidents, and industrial fires. The dataset contains fire-like objects, including yellow or red objects, and sunsets, which make the dataset representative of the challenges that are realistic when addressing fire detection tasks. Notably, the dataset contains both forest and non-forest imagery, which makes it applicable. Examples of this dataset are shown in Fig. 9.

However, to enhance the performance and reliability of wildfire models, the non-forest images were meticulously filtered out during certain phases of model training and testing. BowFire stands out for its versatility in image content and the additional resources it offers. For tasks such as fire segmentation, the dataset includes corresponding masks for fire and non-fire images. These masks provide critical ground



FIGURE 9. BowFire dataset sample images, row 1 shows fire class and row 2 no-fire class. [82].

truth data, facilitating the precise segmentation of fire regions in images. Numerous studies, such as [83], [84], and [85], have used this dataset and achieved meaningful results. Its wide variety of scenarios, combined with the availability of segmentation masks, makes it suitable for developing and testing advanced computer vision models. By simulating real-world complexities and providing high-quality, annotated data, BowFire donates advances in both fire detection and segmentation methodologies.

C. CUSTOM DATASETS

In the period between 2021 and 2024, there has been a large number of works included in the research about wildfire detection, introducing new datasets developed with the help of new methods. Scientists usually use these datasets by obtaining pictures on numerous online sources, including social media, websites, online galleries, wildfire surveillance platforms, and diverse backgrounds of fire scenarios and environmental conditions are represented [77], [86], [87], [88], [89], [90], [91]. However, these datasets often require careful filtering to maintain quality and relevance for training effective fire detection models. For example, Majid et al. [86] introduced a composite dataset constructed by combining images from several well-known public datasets. Chen et al. [77] presented a multi-modal UAV-collected dataset featuring dual-feed side-by-side videos, incorporating both RGB and thermal images, captured during a prescribed fire in an open-canopy pine forest in Northern Arizona. Similarly, Gupta et al. [88] developed the Fire-Flame-Dataset, a comprehensive dataset aggregated from six different sources, designed to train deep learning models for classifying images into fire, smoke, or neutral categories. Mingdi et al. [89] contributed a forest fire dataset consisting of 5,000 images of forest fires, captured at varying scales and in different scenes, to enhance model versatility. In addition to real-world data collection, synthetic datasets have also emerged as a solution to address the challenges of limited data. Park et al. [90] generated synthetic data using Generative Adversarial Networks (GANs) to overcome the limitations of small datasets. Wang et al. [87] advanced this approach by developing the M4SFW (Multi-Faceted Synthetic Dataset for Remote Sensing Forest Wildfire

Detection) using Unreal Engine, allowing simulations across diverse environmental conditions.

Although the publicly available wildfire datasets have been increasing, some of them are geographically and temporally clustered, with a dominant vegetation type, fire regime, and sensing conditions, which might restrict the generalisability of models across regions. In the case of emission-oriented applications, dataset properties that include spatial resolution, frequency of revisit and cloud contamination also affect the accuracy of burned area delineation and fire radiative power retrievals applied in downstream emission modelling. These constraints of the datasets indicate the significance of data diversity and quality in the creation and assessment of wildfire detection and emission estimation frameworks.

VII. COMPUTER VISION TECHNIQUES FOR WILDFIRE DETECTION

Computer vision is a technology that has developed to be used in the detection of wildfires, with numerous image processing and deep learning methods to improve early detection and tracking. In this section, particular attention is paid to the recent research with the aim of analysing various methodologies that have been created to enhance precision and efficiency in detecting wildfires, especially in harsh environments, like dense smoke or the fluctuation of light. This part categorises the mentioned papers according to four major perspectives: High Detection and Segmentation Accuracy, Handling Complex Scenes, Real-Time Detection of Fire and Smoke, and Relying on Multimodal Data. Both schools of thought provide progress and accomplishments in the field of wildfire detection.

A. HIGH DETECTION ACCURACY

Reis et al. [79] propose a novel approach based on transfer learning to detect forest fires using deep CNNs such as InceptionV3, DenseNet121, and ResNet50V2. The paper uses the FLAME dataset which contains high resolution drone images. Using pre-trained models and tuning of hyper-parameters, the approach produces an accuracy of detection up to 99.32% using ImageNet weights in a DenseNet121. Despite its efficiency in terms of the detection speed and accuracy, the computational requirements of this model may be too high for implementation on low-resource devices, thereby limiting the scalability of this model. Further, Kalaivani and Chanthiya et al. [92] reported customised optimized CNN model using Landsat satellite data for forest fire detection. With Antlion Optimization (ALO) being used in the CNN layers, the model will become more accurate and error rates will decrease. The dataset consists of images taken from five forest regions with high performance in fire spread detection. This method is very useful to monitor large forest areas, however it does not provide real-time monitoring because of its high computational demands. In addition, in an attempt to explore CNN architectures, Jonnalagadda and Hashim [93] have used SegNet for real-time segmentation of fire regions from UAV feeds. The result of this model showed

TABLE 3. Overview of computer vision techniques for wildfire detection.

Ref.	Methodology	Dataset	Results	Drawbacks
[22]	Transfer learning using ResNet50, InceptionV3, MobileNet	Multi-source dataset (UAVs, satellites, ground cameras)	Classification accuracy over 99.9%; effective with limited labeled data	Requires extensive data augmentation; challenges in generalizing to new fire patterns
[23]	Deep learning models (EfficientNet-B5, DenseNet201, vision transformers)	UAV aerial imagery datasets	F1-score of 99.9%; effective in small fire region detection within complex backgrounds	High computational requirements; limits realtime application
[79]	Transfer learning using InceptionV3, DenseNet121 & ResNet50V2	FLAME dataset	99.32% accuracy with DenseNet121	limited scalability on low-resource devices
[80]	Reduce-VGGNet & optimized CNN	FLAME dataset	97.35% (optimized CNN); leverages spatial-temporal features	High computational complexity; challenging for real-time deployment
[85]	Learning without Forgetting (LwF) with CNNs (VGG16, Xception)	BowFire dataset	91.41% accuracy (BowFire), 96.89% (original dataset); reduces overfitting	High computational complexity; challenging for resource-constrained applications
[87]	Synthetic dataset (M4SFWD) using Unreal Engine 5	Synthetic dataset with varied weather, lighting, & terrains	Improves model robustness; addresses the lack of large-scale datasets	May not fully capture the complexity of realworld wildfire scenarios
[92]	Custom-optimized CNN with Antlion Optimization (ALO)	Landsat satellite data (5 forest regions)	High accuracy in detecting fire spread	intensive computation limits real-time processing
[93]	SegNet for real-time fire region segmentation	UAV feeds	98.2% accuracy, optimized for realtime use	struggles in low-resolution or adverse weather conditions
[94]	Inception-v3 optimized with RBFNs and RAISR techniques	Curateddatasetforfires. non-fire classification	97.55% accuracy; effective in detecting small fire hotspots	High computational demands due to complex architecture
[95]	Optimization of YOLOv8 using OFAT & random search	Fire & Smoke dataset	Precision improved by 1.39%, recall by 1.48%	Sensitive to environmental noise; prone to false detections in non-fire scenarios
[96]	Self-adaptive algorithm combining physical models with U-Net & YOLOv5	Sentinel-2A, WorldView3	Recall above 81%, reduces false positives	Depends on high-quality images; struggles in low resolution or poor weather
[97]	Optimized YOLOv8 with Ghost Shuffle Convolution and BiFormer attention	UAV-captured images of wildfire smoke	AP of 79.4%; effective in detecting small fires in complex backgrounds	Challenges in processing wooded environments with varying light conditions
[98]	FFIRENet(CNN-based model (MobileNetV2) with IoT integration)	Unique forest fire dataset (urban & forest boundaries)	98.42% accuracy, supports real-time alerts	Dependency on pre-trained layers limits adaptability to varied fire scenarios outside smart cities
[99]	Lightweight CNN model (VIB SD) utilizing multispectral imagery	Landsat imagery (visible, NIR, SWIR; 1,800 images)	High accuracy with reduced computational needs; effective even with partially obscured smoke	Limited to specific spectral bands; may not generalize well to other datasets
[100]	Deep learning model (FireDetXplainer) with XAI tools (Grad-CAM, LIME)	Kaggle & Mendeley public datasets	99.91% accuracy; enhances transparency in wildfire detection	Reliance on pre-trained models; limited adaptability to new datasets without re-training
[101]	Multiscale network (RFWNet) with Digital Twin technology	Synthetic & real-world UAV datasets	High precision across varied terrains; effective with edge computing	High computational complexity limits scalability for large-scale deployments
[102]	Lightweight CNN model (EdgeFireSmoke) for UAV-based detection	UAV video feeds	98.97% accuracy; optimized for realtime edge computing without cloud reliance	Limited performance in low-light conditions; only effective during daytime
[103]	FireXplain-Net (CNN with block enhancements & LIME)	Wildfire Image & FLAME datasets	98.7% accuracy (Wildfire Image), 87.32% (FLAME)	Degrades in complex environments due to simplified architecture
[104]	Multi-scale CNN with residual blocks	Himawari-8 satellite images (1800 samples)	35.2% higher accuracy than traditional methods	Relies on specific satellite data; limits generalization to other regions
[105]	Optimized YOLOv4 model with DenseNet & Spatial Pyramid Pooling	Endangered species data	mAP of 96.89%; effective in challenging conditions	High computational needs; limits real-time applications
[106]	CNN with integration of spatial and temporal data	Himawari-8 satellite imagery (10-minute intervals)	98% accuracy, detects fires 9 mins earlier; reduces false alarms	Reliant on high temporal resolution data; limited effectiveness in areas with persistent cloud cover

98.2% accuracy along with hard computation efficiency, suitable for a real-time application. However, it may suffer from poor performance in low-resolution or bad weather conditions due to its reliance on high-quality input images.

Prakash et al. [94], Yandouzi et al. [22] and Sathishkumar et al. [85] have taken into account the deep learning models and transfer learning methods to improve the fire detection. Prakash et al. [94] have used Inception-v3 based model for detection and classification of biomass burning from forest fires in a CNN based model. The model is trained on a filtered dataset which is a difference between images of fire and images of non-fire. It is trained by Radial Basis Function Networks (RBFNs) and Rapid and Accurate Image Super Resolution (RAISR) methods and has a classification accuracy of 97.55%. The model could be effective, especially in terms of distinguishing the small fire hotspots; however, it is very computationally intensive as the model has a complex structure. On the other hand, Yandouzi et al. [22] employed pre-trained ResNet50, InceptionV3, and MobileNet in their transfer learning experiment. The research utilises a dataset collected through the various sources consisting of UAVs, satellite, and land-based cameras resulting in a precision of classification of above 99.9%. This method is especially useful in cases with scarce labelled data, thus it is very flexible when it comes to quick implementation. But, the fact that it requires large amounts of data to be augmented suggests that generalisation to unobserved fire patterns may be difficult. Meanwhile, Sathishkumar et al. [85] apply the method of transfer learning named Learning without Forgetting (LwF) to enhance the process of fire and smoke detection with the help of the CNNs such as VGG16 and Xception. The system obtained an accuracy 91.41% on BowFire dataset and 96.89% on the original dataset by preserving existing knowledge of the model and acquiring new tasks. In this approach, there are fewer chances of overfitting in line with adaptation to new environments. The resource-constrained applications may be limited, however, by the computational complexity of LwF. The work by Wang et al. is another study where the authors created a deep learning architecture tailored to the detection of wildfires [80]. This framework integrates Reduce-VGGNet for image classification and an optimized CNN model to detect wildfire regions by combining spatial and temporal features. Using the FLAME dataset, the classification model achieved an accuracy of 91.2%, while the optimized CNN reached 97.35%. The study emphasises the use of spatial-temporal data to enhance detection accuracy, particularly for training students in interdisciplinary courses. However, the approach's complexity might pose challenges in real-time deployment due to its high computational requirements.

Another study by Ghali et al. [23] built around UAVs discusses the use of UAVs in conjunction with deep learning models to detect and classify wildfires as well as their segments. The paper employs EfficientNet-B5 and DenseNet-201 as architects, as well as vision transformers,

such as TransUNet and TransFire, to perform segmentation of fire in detail. The method aims at identifying small areas of fire in the complicated backgrounds with an F1-score of 99.9 percent in the aerial imaging datasets. Background complexity and small object detection issues can be solved by the integration of vision transformers, but the computation needs of the technique can make it inapplicable to real-time.

B. HANDLING COMPLEX SCENES

Recent research has focused on optimizing YOLO-based architectures to enhance the accuracy and adaptability of wildfire and smoke detection across diverse environments. Ramos et al. [95] optimized the YOLOv8 model using the Fire and Smoke dataset for agricultural and environmental monitoring. By employing the One Factor at a Time (OFAT) and random search methods, the model improved precision by 1.39% and recall by 1.48%, though it remained sensitive to environmental noise that can cause false detections. Jin et al. [96] developed a self-adaptive framework combining physical modeling with deep learning, integrating U-Net for segmentation and YOLOv5 for false-positive filtering. Using Sentinel-2A and WorldView-3 imagery, their model achieved a recall rate above 81% and effectively reduced false alarms in complex terrains, albeit with high computational demands. Similarly, Saydirasulovich et al. [36] improved YOLOv8 for UAV-based smoke detection by integrating Ghost Shuffle Convolution layers and BiFormer attention mechanisms. The average precision of the model with UAV imagery was 79.4%, which is higher in detecting the small plumes of smoke, but the ability is lower on heavily wooded or changing light situations.

Complex scenarios are also dealt with by CNN-based architecture, such as Khana et al. [97] introduced FFireNet, a CNN-based model to perform early-stage fire detection in smart city scenes on the MobileNetV2 architecture. The study runs pre-trained models on a new dataset of fully connected layers to make forecasts, which gives a mean accuracy of 98.42% in a discrete forest fire dataset, most specifically, in urban and in the boundary of a forest area. IoT and real-time video surveillance will give alerts to the authorities in time. Nevertheless, the model can be less adaptable to the variety of fire conditions beyond the cases considered in the smart city since the model relies on already trained layers. Additionally, Zhao et al. [98] trained a lightweight CNN network named VIB_SD which identifies fire smoke based on multispectral (Landsat) images that consist of visible and infrared (IR) images. This paper emphasizes the advantages of employing the Near-Infrared (NIR) and Short-Wave Infrared (SWIR) Bands that enhance the efficiency of detection particularly when the smoke is partially covered by clouds. The dataset used in the study comprises over 1,800 images, categorised into “Smoke,” “Clear,” and “Other Aerosol.” The model achieved high accuracy with reduced computational requirements, making it suitable for small satellite applications.

A study by Rubab et al. [99] on Explainable AI (XAI) proposed a deep learning neural network combination of MobileNetV3 and the XAI methods Grad-CAM and LIME (for transparent wildfire detection). The model employs transfer learning with a classification accuracy of 99.91% on datasets obtained at kaggle and Mendeley. Through the use of XAI, the model will be more transparent, and it will assist the stakeholders to understand the decision-making process, and this can significantly enhance emergency management. Nevertheless, being extremely precise, there is an issue of their lack of adaptability to new data unless retrained extensively.

C. REAL-TIME DETECTION OF FIRE AND SMOKE

For real-time detection, researchers like Wang et al. [100] propose RFWNet, a multiscale network leveraging Digital Twin (DT) technology for real-time forest fire detection using UAVs and edge computing devices. The network uses deformable convolutions and dynamic sparse attention mechanisms to enhance detection in diverse terrains. It combines synthetic datasets and real-life data to provide training to the model demonstrating high accuracy in different conditions. Although RFWNet has better accuracy, it is more intricate, which raises the cost of computation, which may make it difficult to scale to large systems. On the same note, Almeida et al. [101] introduced EdgeFireSmoke a light remote sensor file and edge computing CNN optimized to detect fire and smoke in real time with UAVs and edge computing, customised to meet the needs of small and energy-constrained devices that frequently operate on batteries and might be deployed remotely akin to iCare at any time. The model gets the video feeds of drones to identify fire and smoke with 98.97 percent precision. It is also capable of working on edge devices, which does not require cloud computing and this is an advantage to areas that are remote and inaccessible. However, the model performs only in the daytime, as it is not good in working in low lights.

Additionally, Khan and Park [102] proposed FireXplain-Net, a CNN model which was optimized on the wildfire detection through the use of convolution block architecture improvements. On the Wildfire Image dataset and FLAME dataset, the model was found to be truly impressive with an accuracy of 98.7% and 87.32%, respectively. The development of this work is aimed at the optimization of the accuracy and the computational efficiency, which is why it is appropriate to real-time application. The integration of Local Interpretable Model-Agnostic Explanations (LIME) further improves model transparency. However, the model's performance may degrade in highly complex environmental conditions due to its relatively simplified architecture.

Hong et al. [103] in another study propose FireCNN, that employs multi-scale convolutional layer with residual blocks to identify active fires in Himawari-8 satellite imagery. The model also overcomes the drawbacks of the traditional threshold-based approach, as the model can detect with an accuracy of 35.2% greater than the traditional threshold

methods. It has more than 1,800 fire and non-fire samples, with an interest in the Asia-Pacific region. There is the possibility of near real time monitoring of fires due to the use of high temporal resolution satellite data but the model may be restricted in use by other areas due to its reliance on specific satellite data. In addition to that, Roy et al. [104] suggest WildDect-YOLO, an optimized model that is based on the YOLOv4 model to detect wildlife in real-time use-case scenarios. The DenseNet and Spatial Pyramid Pooling enhancements improve the feature extraction with a high mAP of 96.89. The model performs well under rigid conditions and has high processing requirements tested on a dataset of endangered species. This study highlights the importance of supranational sensors in the preservation of wild animals, where human participation is not necessary.

D. DEPENDENCE ON MULTIMODAL DATA

The study by Kang et al. [105] presented a deep learning model using data from the Himawari-8 geostationary satellite to detect forest fires with minimal latency. This study combines CNNs with spatial and temporal data to enhance detection accuracy. By using high-resolution images captured at 10-minute intervals, the model demonstrated an ability to detect fires up to 9 minutes earlier than existing methods, achieving an accuracy of 98%. The strength of this model lies in its ability to integrate temporal information, particularly reducing false alarms. However, its reliance on high temporal resolution data may not be applicable in regions with persistent cloud cover. Also, Almasoud [106] introduced a novel approach by combining an Attention-based CNNs with a Bidirectional Long Short-Term Memory (BLSTM) network. This hybrid model utilised a Bacterial Foraging Optimization (BFO) algorithm for hyperparameter tuning, which enhanced its accuracy in detecting wildfire signatures. The model was tested on a comprehensive multisensory dataset, which included thermal, optical, and multispectral data collected from a forest monitoring setup. The ACNN-BLSTM model achieved an impressive accuracy of 99.56%, demonstrating robust performance in identifying early fire signals across various conditions. However, the study highlighted computational challenges due to the model's complex architecture, which involves several layers and the additional computational load of BFO optimization. While highly effective, the system's reliance on sensor networks may limit its deployment in remote areas without communication infrastructure.

Seydi et al. [107] developed Fire-Net, a deep learning model for active fire detection using Landsat-8 imagery that combines optical (RGB) and thermal bands. By adding the residual convolutional blocks the model was able to achieve an accuracy of 97.35 percent at a variety of different regions including North America, the Amazon, and Chernobyl. Fire-Net is effective to detect small spots of fire, it uses high-resolution satellite, which makes this model not very effective in places with low coverage. In a study by Wang et al. [87], the data scarcity was solved by the

creation of the synthetic M4SFWD dataset by using the Unreal Engine 5, in which the differences in weather, light, and terrain were simulated. Synthetic data improves the robustness of models and increases the scalability of training, but it could fail to accurately represent the complexity of scenarios of wildfires in reality.

Abdusalomov et al. [108] developed the improved version of forest fire detection with the help of the Detectron2 model and a new dataset of 5,200 images, taken under different light conditions. The dataset was enhanced by massive data augmentation to 119, 600 images and this allowed the model to attain a precision of 99.3 percent. Although it is very precise, the model is computationally costly, it cannot be implemented in real-time in the environment with low resources. Rahman et al. [110] created a UAV-supported method of wildfire smoke detection based on the texture and color analysis of UAV images with the micro-average F1-score of 0.87. The technique is good in controlled environments but it is limited to low-light and dense canopy conditions. MultiScale-Net was active fire detector model by Rostami et al. [110] that is a deep multi-kernel learning on Landsat-8 image. It has an F1-score of 91.62, which effectively traces the small and isolated fire spots (using the Active Fire Index (AFI) of SWIR and Blue bands). Nevertheless, it can be limited in use in regions with no such data because of its reliance on particular spectral bands.

Buza and Akagic [55] and Wang et al. [80] both worked on the best approaches to improving the detection of wildfires and how available data can be used in different ways. The article by Buza and Akagic discusses an original unsupervised method of fire and wildfire segmentation and detection in the RGB colour space only [55]. This technique also does not need any time-intensive conversion of colour space, unlike in the previous method that requires use of the HSV or YCbCr colour space. This model uses image processing to gain flame features directly with an improvement of F1 score of 3% compared to the current methods. The dataset has still photos in different settings and in this way, the model will effectively identify flames even where the light situation is fluctuating. Nonetheless, this model can only be effective, and its use on dynamic wildfires in real time might be constrained by the use of static images.

E. ARCHITECTURAL REPRESENTATION OF DEEP LEARNING MODELS

To enhance clarity and reproducibility, the deep learning models surveyed in this work are described in terms of their input layers, hidden layers, and output layers, following established deep learning design conventions and recent wildfire detection studies [36], [37], [38].

1) INPUT LAYER

The input layer receives visual data derived from RGB images, thermal infrared imagery, multispectral satellite data, or fused multimodal inputs. UAV-based approaches commonly process high-resolution RGB or RGB-thermal

images (e.g., 640×640 pixels) for early fire detection [36], [38] while satellite-based studies utilise multispectral bands from platforms such as Landsat and Sentinel-2 to capture large-scale fire activity. In multimodal architectures, separate input branches are employed to process RGB and thermal channels independently before fusion, improving robustness under low-visibility and nighttime conditions [92], [96].

2) HIDDEN LAYERS (FEATURE EXTRACTION AND REPRESENTATION)

The hidden layers form the core of wildfire detection models and are responsible for hierarchical feature extraction. CNN-based approaches, such as DenseNet121 and SegNet, employ stacked convolutional layers to capture low-level features (edges, textures) and high-level semantic patterns (smoke plumes and flame contours) [79], [93].

YOLO-based architectures utilise a backbone network (e.g., CSPDarknet, MobileNet, or EfficientNet) followed by feature aggregation modules such as Feature Pyramid Networks (FPN) or Path Aggregation Networks (PAN) to enhance multi-scale detection of small and distant fires [37], [38]. Recent improvements incorporate attention mechanisms, including CBAM and BiFormer attention, to refine feature representations and reduce false positives in complex forest backgrounds [36].

Transformer-based models further enhance hidden-layer representations by introducing self-attention mechanisms that capture long-range spatial dependencies. These architectures are particularly effective in detecting irregular fire boundaries and smoke regions under cluttered or occluded conditions, as demonstrated by Swin Transformer-based segmentation models [46].

3) OUTPUT LAYER

The output layer is task-specific and depends on the wildfire monitoring objective. For classification tasks, the output consists of binary or multi-class probability scores indicating fire or non-fire presence. For object detection, models such as YOLOv7 and YOLOv8 produce bounding box coordinates, confidence scores, and class labels for fire and smoke regions [37], [38], [41].

In segmentation-based approaches, pixel-wise probability maps are generated to delineate fire fronts and smoke plumes, supporting precise fire perimeter mapping and downstream emission estimation workflows [45], [46]. These outputs are further post-processed to support real-time alerts, burned-area estimation, and integration with emission models such as GFAS.

The use of computer vision algorithms has consequently greatly enhanced the process of detecting and tracking wildfires at an earlier stage. The latest models which include CNNs, Mask R-CNN, and YOLOv8 especially when implemented on UAV imagery provide high precision and real-time reactivity even in the complex environmental conditions. It has been demonstrated that these methods improve precision and effectiveness of the firefighting process, yet

the processing of large-scale data, smoke separation, and real-time flexibility are still difficult. Table 3 gives a summary of available techniques, datasets, results and limitations. The future study should be aimed at enhancing generalization and computational efficiency of the models. In general, the analyzed literature sheds light on a high potential of a combination of computer-vision and the conventional methods of remote-sensing to manage forest fires successfully. Nevertheless, the comparison of the performance using heterogeneous data and assessment processes makes the interpretation of a direct comparison of the models, which obligates the development of benchmark datasets and standardized validation models. While reported accuracy and mAP values provide useful performance indicators, direct comparison across wildfire detection models remains challenging due to variations in datasets, sensing platforms, and evaluation protocols. Beyond accuracy, CNN-based models often exhibit higher robustness under controlled conditions but face scalability challenges for real-time deployment, whereas YOLO-based architectures prioritise inference speed and edge compatibility at the expense of fine-grained segmentation accuracy. Transformer-based models demonstrate improved contextual reasoning and boundary delineation but typically incur higher computational costs, limiting their applicability in resource-constrained environments. These trade-offs show the importance of selecting models based on operational constraints rather than accuracy metrics alone.

VIII. STUDY OF WILDFIRES EMISSIONS QUANTIFICATION

Wildfire emissions are now understood to be the primary source of atmospheric pollutants, contributing to climate change and affecting human health and ecological systems. Such emissions mostly include carbon dioxide (CO₂), particulate matter (PM_{2.5}), and nitrogen oxides (NO_x) and other pollutants and could travel long distances and reach areas that are very far and distant to the source of the fire. The Copernicus Atmosphere Monitoring Service (CAMS), via its GFAS, tracks wildfire emissions worldwide using satellite data. As shown in Fig 10, annual wildfire carbon emissions have fluctuated from 2003 to 2023, with significant peaks during years of intense fire activity. The recent spike in 2023 is largely due to severe fires in the Canadian boreal forests, driven by extreme heat and prolonged drought. These emissions degrade air quality and contribute to global carbon levels, highlighting the need for better wildfire management strategies. Recent studies have explored the scope of wildfire emissions, focusing on their spatial distribution, intensity, and the resulting environmental impacts.

In the study conducted by Jones et al. [3], it was found that wildfires in 2023–2024 led to a total of 2.4 Pg of carbon emissions globally, driven largely by unprecedented fire activities in the Canadian boreal forests, which alone recorded emissions nine times higher than the historical average. This intensification of fire regimes, particularly in boreal regions, is linked to rising temperatures and extended drought periods due to climate change, which depletes soil

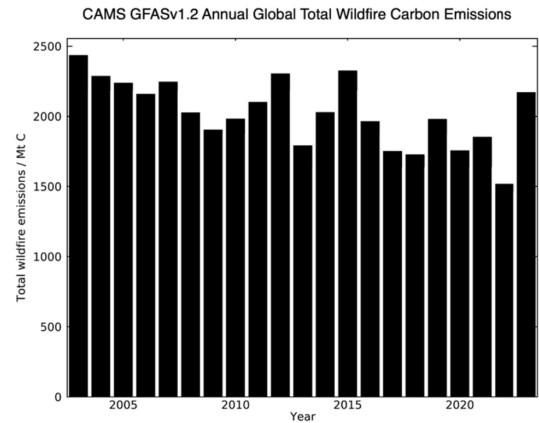


FIGURE 10. Annual global total estimated wildfire carbon emissions from 2003 to 2023. Source: CAMS.

and vegetation moisture, making these areas more fire-prone. In Canada, MacCarthy et al. [111] documented the extensive impact of the 2023 wildfire season, where nearly 7.8 million hectares of forest were burned. These fires accounted for over 25% of global tree cover loss and released approximately 3 billion tons of CO₂, exceeding emissions from tropical deforestation. The extreme heat and low precipitation in 2023 were key factors driving these unprecedented fires, exacerbating the climate crisis by turning forests from carbon sinks into sources. Further emphasising the severity of recent wildfires, Byrne et al. [112] utilised satellite-based inverse modelling to quantify the emissions from the 2023 Canadian fires. Their findings revealed that these fires emitted approximately 647 teragrams of carbon (TgC), comparable to the annual fossil fuel emissions of countries like India. The study attributed the surge in fire activity to unusually hot and dry weather conditions, which are projected to become more common by the 2050s under moderate climate change scenarios.

Xu et al. [113] examined rising emissions from Arctic wildfires, particularly in continuous permafrost regions. The study reported that wildfire emissions from these areas have grown from 2.42% of global emissions in 1997 to over 20.86% in 2021. This dramatic increase is attributed to the thawing of permafrost, which releases ancient carbon stores, thereby accelerating the positive feedback loop of global warming.

Smith et al. [121] highlighted that forest fires in the United States, particularly in Alaska and the western states, have been increasing both in intensity and frequency. The Wildland Fire Emissions Inventory System (WFEIS) reported emissions of 129 Tg of CO₂ in 2022, emphasising the growing contribution of wildfires to national greenhouse gas inventories. This study noted that while forests act as carbon sinks, large-scale wildfires can offset much of this carbon storage, leading to higher levels of CO₂ in the atmosphere.

TABLE 4. Summary of studies on wildfire emissions and impacts.

Ref.	Methodology	Key Findings	Region
[3]	Burned Area and Fire Radiative Power (FRP) anomaly detection from Earth observations	Global wildfires in 2023-2024 led to 2.4 Pg of carbon emissions; Canadian boreal forests had 9 times higher emissions than averages.	Global, Canada
[111]	Satellite-derived tree-cover loss combined with forest carbon-flux modelling to quantify stand-replacing fire extent and CO ₂ emissions.	7.8 million hectares burned in Canada, contributing to over 25% of global tree cover loss and releasing 3 billion tons of CO ₂ .	Canada
[112]	4D-Var inverse model in CMS-Flux assimilated TROPOMI and MOPITT CO data against GFED, GFAS, and QFED fire inventories, then converted CO to CO ₂ using emission factors.	Canadian fires emitted 647 Tg of carbon, comparable to India's annual fossil fuel emissions, due to hot and dry weather.	Canada
[113]	Mapped GFAS CO ₂ emissions to permafrost zones and used 1997–2021 trends with ERA5 soil moisture to assess drying-driven belowground combustion.	Arctic wildfire emissions rose from 2.42% (1997) to over 20.86% (2021) due to permafrost thaw releasing ancient carbon stores.	Arctic
[114]	Used WFEIS to combine GIS fire perimeters, CONSUME fuel-consumption outputs, and vegetation-specific emission factors to estimate CO ₂ emissions.	U.S. wildfires emitted 129 Tg of CO ₂ in 2022; offset of carbon sequestration from forests.	United States
[115]	Reconciled multiple fire-information sources, satellite active-fire detections, plus national, state, tribal, and local fire reports into a single wildfire emissions inventory using the CFIRE framework.	Developed methodology for integrating satellite data with regional reports to improve emission estimates.	United States
[116]	Applied GFED4.1s to merge MODIS burned-area and FRP data with fuel-consumption factors, then correlated 2003–2019 emissions with ERA5 temperature anomalies.	23% of U.S. PM _{2.5} concentrations in 2020 from wildfires; over 3,720 NAAQS exceedances.	United States
[117]	Activity-based GHG accounting using the GHG Protocol and DEFRA emission factors, combining project energy use, commuting, materials, and event data, with scenario analysis for mitigation	83% of nitrogen emissions in the western U.S. during the 2020 fire season, increasing nitrogen deposition by 78%.	United States
[118]	Overlaid Landsat burn maps on Sentinel-2–derived biomass carbon layers to calculate fuel consumption and estimate CO ₂ emissions using standard emission factors.	Wildfire PM _{2.5} in Brazil linked to a 1.47% rise in preterm births (2010-2019).	Brazil
[119]	Reviews ground and satellite pollutant measurements, dispersion modelling, and ecological impact studies to identify knowledge gaps in wildfire-derived air pollution risk.	Wildfire pollutants can travel vast distances, affecting soil and water quality.	Global
[120]	Simulated wildfire and two prescribed-burn scenarios in the CMAQ air-quality model using scenario-specific emissions, then compared resulting PM _{2.5} concentrations and health impacts against EPA monitoring data.	Prescribed burns reduce PM _{2.5} emissions by up to 52% compared to wildfires.	California, United States
[121]	Collected airborne measurements of trace-gas and particulate emission rates over western U.S. wildfires and compared them against remote-sensing–based emission inventories for validation.	129 Tg of CO ₂ in 2022, emphasising the growing contribution of wildfires to national greenhouse gas inventories.	United States
[122]	Instrumented research aircraft directly sampled wildfire smoke over western U.S. fires and compared those in-situ measurements against satellite-derived and inventory-based emission estimates to validate their accuracy	SERA database compiles emission factors for various pollutants in North American wildfires.	North America
[123]	MODIS-derived fire radiative power pixels were classified by intensity thresholds to estimate Siberian wildfire carbon emissions.	Siberian wildfires could emit up to 1500 Tg C annually by 2050 if current patterns persist.	Siberia
[124]	Used supersite trace-gas and particle measurements with Lagrangian modelling to derive emission factors and validate against GFAS, GFED, and FINN inventories.	Small-scale wildfires impact regional air quality, increasing CO and CH ₄ levels up to 200 km away.	France
[125]	TROPOMI CO, MERRA-2 black carbon, and AIRS humidity products were processed in Google Earth Engine and QGIS to map pollutant distributions and assess meteorological influences on wildfires	Meteorological conditions affect wildfire emissions; pollutants can reach altitudes of 14 km and impact the global atmosphere.	USA, Brazil, Australia

In the Comprehensive Fire Information Reconciled Emissions (CFIRE) inventory, Larkin et al. [115] developed a methodology that integrates multiple sources, including satellite data and regional fire activity reports, to produce more accurate emissions estimates for the U.S. Environmental Protection Agency's National Emissions Inventory for 2011 and 2014. This inventory emphasised the importance of including data on smaller and prescribed fires, which are often missed in satellite-only datasets.

Li et al. [116] investigated the impact of the 2020 wildfire season in the western United States, which was one of the most severe on record. The study found that wildfires were responsible for 23% of surface PM_{2.5} concentrations across the contiguous United States from August to October 2020. In particular, the Pacific Coast and Mountain regions saw even higher contributions of 43% and 42%, respectively. The fires resulted in over 3,720 exceedances of the National

Ambient Air Quality Standards (NAAQS), affecting public health. Similarly, the historic 2020 wildfires led to nitrogen emissions, as noted by Campbell et al. [117]. These fires were responsible for up to 83% of the total nitrogen emissions in the region during the fire season, leading to a 78% increase in nitrogen deposition in California's forests. The nitrogen deposition rates were found to be up to 12 times the critical load threshold, which can lead to detrimental effects on forest ecosystems and biodiversity.

Health impacts were also studied by Zhang et al. [118]; they conducted a nationwide cohort study in Brazil. The study estimated that wildfire PM_{2.5} contributed to a 1.47% increase in preterm births from 2010 to 2019, particularly in low-income regions. Paul et al. [119] examined the downwind effects of wildfire emissions, finding that pollutants like PM_{2.5} and volatile organic compounds (VOCs) can travel vast distances and deposit harmful substances in ecosystems, affecting soil and water quality.

Kiely et al. [120] examined the comparative emissions of wildfires versus prescribed burns in California. The study demonstrated that prescribed burns could reduce PM_{2.5} emissions by up to 52% compared to uncontrolled wildfires. This reduction in emissions underscores the potential of prescribed burns to manage forest fuels, reduce wildfire severity, and limit associated health impacts.

The study by Stockwell et al. [124] validated the use of remote sensing for measuring wildfire emission rates, demonstrating strong correlations between aircraft-derived emission rates and satellite data. Integrating in-situ measurements with satellite observations provided more accurate estimates of carbon and other trace gas emissions from wildfires. This method enhances the accuracy of emission inventories, supporting more effective air quality management and reliable climate projections. The Smoke Emissions Repository Application (SERA) database, presented by Prichard et al. [122], compiled extensive emission factors (EFs) for pollutants like CO, CO₂, CH₄, and PM_{2.5} across North American wildland fires. This repository synthesises data from multiple sources, including prescribed burns and wildfires, providing useful insights into the variability of emissions based on combustion phase, fuel type, and geographic region.

The 2020 wildfires in Siberia, studied by Ponomarev et al. [123], highlighted the alarming trend of increasing carbon emissions ranging from 20 to 350 Tg of carbon annually. The study underscored that under extreme climate scenarios, emissions could reach up to 1500 Tg C by 2050 if current vegetation patterns and fire management strategies remain unchanged. This increase in emissions is driven by higher fire intensity, particularly in boreal forests where deep-burning fires release large amounts of stored carbon from peatlands.

In a case study from France, Xue et al. [114] investigated the regional air quality impacts of a wildfire near Orléans. They found that even small-scale wildfires can noticeably affect regional air quality, with increased CO, CH₄, and black carbon concentrations observed up to 200 km from the burn site. This study underlined the discrepancies in emission estimates from different models, highlighting the need for better inventory systems to capture the emissions from smaller wildfires accurately.

The research by Shikwambana and Kganyago [125] focused on the 2018–2019 wildfires in the USA, Brazil, and Australia, which demonstrated that meteorological conditions such as temperature and humidity influence wildfire emissions. Using satellite instruments like CALIOP and TROPOMI, the study mapped the vertical distribution of smoke and found that emissions from Brazil reached altitudes of up to 14 km, allowing pollutants to travel vast distances and affect atmospheric conditions globally.

These studies highlight the urgent need for improved wildfire management strategies, such as prescribed burns, early detection systems, and more accurate emissions inventories. By integrating satellite data, ground measurements, and advanced modelling techniques, researchers can better

inform policy decisions to mitigate the adverse effects of wildfires on ecosystems, air quality, and human health. Unlike wildfire detection models, which are driven by architectural choices in deep learning, emission quantification frameworks primarily rely on detection-derived inputs such as burned area maps and fire radiative power (FRP), combined with fuel load estimates, combustion completeness, and emission factors. Consequently, technical differentiation across emission models arises less from learning architecture and more from input resolution, temporal aggregation strategies, and parameterisation assumptions. This distinction explains the comparatively lower architectural diversity in emission modelling while underscoring its dependence on accurate and timely detection outputs. The overall summary of each study, including the study region, is given in Tab. 4. A persistent challenge in linking wildfire detection and emission quantification is the discontinuity between the accuracy of detected fire perimeters and the precision of derived emission estimates. Since emission modelling pipelines rely on detected burn-area masks and fire radiative power retrievals, any misclassification or spatial uncertainty in the detection stage propagates into emission calculations. Moreover, the models of emission estimation are sensitive to uncertainties in upstream detection output. The inaccuracies in burned-area mapping, fire perimeter mapping and fire radiative power estimation are carried over to the emission calculations together with uncertainties in fuel-load characterization, full combustion and emission factors. Uncertainties are also caused by variations in sensor resolution, time sampling, and the atmosphere. Although the present review does not present any novel mathematical equations, it demonstrates that more studies are required in future to mathematically represent and propagate any uncertainties in the detection stage with the use of emission models to enhance robustness and comparability between regions. Future research should focus on developing integrated frameworks that jointly optimize detection and emission estimation, incorporating uncertainty propagation across stages to improve the reliability of regional and global emission inventories. Future research should therefore prioritise the development of integrated detection–emission frameworks that jointly optimize fire localisation and emission estimation, explicitly incorporating uncertainty propagation across pipeline stages. Such approaches would improve the robustness and reliability of regional and global wildfire emission inventories, particularly in heterogeneous landscapes and under complex atmospheric conditions.

IX. DISCUSSION

The integration of computer vision and remote sensing technologies has greatly advanced wildfire monitoring and response. Deep learning models, particularly CNN-based approaches, have proven highly effective in detecting fire and smoke across diverse data sources such as satellite imagery, UAV footage, and thermal infrared data. These innovations enable real-time tracking and faster response,

minimizing the damage to ecosystems, infrastructure, and human life. By combining high-resolution imaging with powerful machine learning algorithms, detection systems now achieve greater precision and speed, ensuring timely intervention during wildfire events.

Our survey shows that although much progress has been made in both wildfire detection and emissions modelling as two separate areas, relatively little research links these two areas together in an end-to-end evaluation. Integrating detector outputs with emission quantifiers not only enables near-real-time impact assessment but also offers a feedback mechanism to refine detection algorithms using observed emission discrepancies. For example, spatial biases in detection masks can propagate into over- or under-estimations of carbon release, highlighting the need for joint calibration protocols. Future work can therefore focus on co-optimizing detection accuracy and emission-model fidelity, potentially via differentiable assimilation layers or unified loss functions, to close the loop between early warning and environmental accounting.

A. SCALABILITY

However, even though significant advancements have been achieved in this area, there are still a few issues that should be tackled. Scalability of systems based on the high-resolution satellite imagery and UAV data is one of the main challenges. Although satellites such as Himawari-8 and Sentinel-2 can offer a comprehensive coverage and useful data, it may not be feasible to introduce such technologies in remote or underdeveloped areas with little technological infrastructure. The computational overheads incurred in the processing of large scale high-resolution images render real time applications difficult, particularly in resource limited environments where the availability of sophisticated computing resources may be restricted.

B. ENVIRONMENTAL VARIABILITY

Another significant issue of wildfire detection is demonstrated by the variability of environmental conditions. Wildfires frequently happen in the place with a lot of smoke in the atmosphere, in the clouds, or when the light is low, and this may diminish the usefulness of detection algorithms that use mostly visible light (RGB) images. Although these limitations can be partly addressed by thermal sensors of infrared and multispectral imaging, they add additional complexity and cost. These sensors are better suited to low visibility settings and can sense heat signatures but are not consistently easily accessible and cost effective to use in large numbers. This has created the necessity of more cost effective and scalable solutions which can work well under different environmental conditions.

C. DATA FUSION

In addition, merging information of various sensors, including satellites, UAVs, and ground-based systems, is not a technically simple task. It is important that sensor fusion

is essential to build a strong wildfire monitoring system, but this will demand advanced algorithms that can handle a mixed stream of data in real-time. Integrating the information of various sources contributes to eliminating the limitations of separate systems and providing a comprehensive and trustworthy picture of fire incidents. Nevertheless, the creation of algorithms capable of executing such an integration effectively and with the lowest amount of latency is still being researched.

D. SYNTHETIC DATA

Synthetic datasets and transfer learning have been shown to be effective in training models in situations where labelled data is limited, however such a technique has several drawbacks. Although artificial data created by software such as Unreal Engine 5 can improve model training, offering a more diverse set of scenarios, it might not reproduce complex dynamics of wildfire conditions in real life in all aspects. This restriction may cause problems of model generalisation, in which systems do not work in uncontrolled or real-world contexts (that is, in the real wildfire event). Thus, there is an increasing interest in exploring more realistic, real-world data to be able to make the models more robust and operate with a higher likelihood of success in the practice.

E. EDGE DEPLOYABLE MODELS

Besides these issues, the research studies analysed in this paper reveal that it is necessary to develop lightweight and efficient models that can be deployed to edge devices. The transition to edge computing provides a good resolution to the low-latency issue since the data can be processed at the point of collection on drones or field devices and that way automatic dependence on cloud infrastructure is minimised. As an example, models such as EdgeFireSmoke have shown that it can be possible to get high accuracy, yet cut computational load. These models are particularly useful in far locations where the connectivity can be less than ideal, which means that the wildfire fires can be detected and reacted to more quickly. Using algorithms optimized to perform edge computing might allow new areas that could not have detection systems before because of technological constraints to be deployed. However, most edge-deployable approaches reported in the literature have so far been evaluated under experimental or pilot-scale conditions, highlighting the need for broader validation within operational wildfire management systems.

F. OPTIMIZATIONS

In addition, there is an immediate need to increase the resilience and adaptability of such systems of detecting wildfires as the frequency and severity of wildfires increase with a change in climate. Future studies may be aimed at creating models that are both resource-efficient and correct in their application to be able to work in a wide variety of environments with limited computational power. Pruning, quantisation and specialised hardware accelerator

development are a few of the techniques that can be considered to optimize models used in real-time on low-power machines. This is especially vital in areas where time and again the need to detect fire at the earliest is paramount to avoid any spread of the fires and result in the devastation of property.

G. LIMITATIONS OF DEEP LEARNING-BASED FIRE DETECTION

Although deep learning-based methods of wildfire detection have a high accuracy, they have several significant limitations:

1) DATA DEPENDENCY AND GENERALIZATION

Most do need big and well-marked datasets; they tend to fit to the circumstances (type of terrain, intensity of fire) in their training data and they fail to extrapolate to new areas or sensor types.

2) ENVIRONMENTAL ROBUSTNESS

Under heavy haze and cloud cover, low-light or night conditions where performance worsens, and when models are trained on clear RGB images, they cannot be depended upon to discriminate fire and background noise.

3) COMPUTATIONAL COSTS

State of the art CNN and transformer models are resource intensive in terms of compute and memory during training and inference, and are currently not deployable on edge devices without vigorous pruning, quantisation, or custom accelerators.

4) DOMAIN SHIFT AND TRANSFERABILITY

Models that are trained on the vegetation types or climate of a single region tend to lose accuracy when applied to a different region (e.g. boreal vs. Mediterranean forests), and will need expensive domain adaptation or retraining.

5) INTERPRETABILITY AND TRUST

Deep networks are black-box, so failure modes are difficult to detect and why a model failed to detect a fire or raise false alarms is challenging, which negatively affects the trust of operators and makes them hard to integrate into actual operations.

Overcoming these drawbacks will be necessary to transition between the proof-of-concept research to systems that are robust and scalable to support year-round and global wildfire monitoring.

The developments in AI and deep learning provide more proactive opportunities in management of wildfires. Predictive analytics, which is applied with the help of historical information and real-time monitoring, might enhance the possibility to predict fire danger and propagation. Both these abilities, in combination with an early warning system, will be able to make firefighting operations more effective, manage resources in a more effective way, and eventually safeguard

natural ecosystems, as well as human settlements. These future directions and the need to tackle the current constraints will play an important role in the development of overall wildfire management systems that will be able to sustain the increasingly threatening climate change. Generally, this survey emphasizes that the stability of the wildfire emission estimation is closely correlated with upstream detection and segmentation performance, and thus it is necessary to consider integrated and uncertainty-sensitive pipelines to explicitly consider the data bias, sensor performance and deployment constraints to help translate computer vision improvements into operational and policy-relevant wildfire monitoring systems. Further, the literature survey reveals that certain areas and ecosystems are geographically concentrated and especially in places where the satellite coverage is established or datasets are publicly available. Because of this, several detection models are trained and tested on particular vegetation types, climatic conditions and fire regimes, and might not be generalizable. The downstream impact of this bias on the quantification of emissions is that fuel properties and combustion processes differ significantly among ecosystems. The solution to these gaps will involve the creation of a greater number of geographically diverse datasets and cross-region validation.

X. CONCLUSION

The current review mentions the game-changing role of computer vision and remote sensing technologies in the development of wildfire detection and emissions monitoring. The accuracy and speed of fire detection have been enhanced by the creation and optimization of deep learning models, including CNNs, YOLO models, and hybrid systems that combine transformers and attention mechanisms. These developments are critical, particularly with the ongoing patterns of climate change which are fueling the occurrence of more and more intense wildfires on earth.

Nevertheless, the review research also shows that there are still ongoing challenges that should be overcome. Among the lessons learned, one should note that accuracy of detection is important at the expense of computer efficiency. Although most of the state-of-the-art models are very accurate, they need a large amount of computation and thus they can not be practically applied in real time applications. Solving such issues will involve the development of lighter models that would run on edge devices without compromising the performance. The importance of this need is particularly relevant to low-resource or climate-prone areas where low-cost, modular infrastructure can facilitate the early warning needs.

Further, integrating multiple data sources, including satellite imagery, UAV feeds, and ground-based sensors, holds great promise for improving the robustness of wildfire detection systems. Multi-modal approach can reduce the limitations of any single source of data hence improving the overall reliability of the fire detection particularly in challenging conditions. Also, more complex sensor fusion,

along with the prospect of Digital Twins, will be able to allow real-time simulations and forecast fire propagation in order to inform decisions in fire fighting activities. Combining these detection capabilities with emissions estimation, using fire radiative power and burn-area maps, can further support dynamic environmental monitoring and response. However, the scalability of UAVs and edge-computing systems for emission monitoring remains limited, emphasising the need to evaluate their practical and economic viability for large-scale operational deployment.

Future studies need to combine UAV and ground-based computer vision data as high-resolution validation data to enhance satellite-based fire and emissions models to maintain consistency between the local precision of detection at the scale of local datasets and global-scale monitoring systems. The future research also needs to focus on creating open and standardised wildfire data and assessment systems to facilitate significant cross-model comparison. The next challenge in closing the continuity gap between accuracy of detection and emission estimation is still a major area of future work and this would need the incorporation of models that would propagate uncertainty in visual detection to quantification of atmospheric impact. Further research is also needed to streamline models to be scaled and used in the real-time. It will be important to develop algorithms that can be executed on low-power devices, to combine data obtained by different sources of imaging, and to use machine learning models that can adapt to various environmental conditions. Generative models like GANs might also improve model generalisation by synthesising wildfire situations to train models, particularly in infrequent or difficult to capture situations, including during the night or heavy smoke. With the wildfires becoming more frequent and severe, these technological innovations will come in handy to reduce the consequences, preserve the ecosystems, and secure communities.

Through further expanding the limits of what could be achieved through computer vision and remote sensing and AI, we can create more robust systems that will be able to cope with the complexities of wildfires. Finally, it is not only about detecting fire but also about delivering practical insights that can be used in the development of effective response strategies, thus minimising the harmful impact of wildfires on human life and the environment.

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